Bonds Age, Pricing Errors Persist*

Sunho Lee†

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Abstract

Pricing errors in term-structure models are often strongly autocorrelated. In a preferred-habitat model, I show analytically that the aging of issued bonds can generate residuals with small variances and strong autocorrelation. Bonds issued at one date remain outstanding at slightly shorter maturities at the next. The same issuance therefore affects nearby points on the yield curve at successive dates, creating autocorrelation even when new issuance is independent. I call this the issuance cohort effect. In the calibrated model, removing the cohort effect reduces the autocorrelations of higher principal components of yields to nearly zero. Thus, bond pricing residuals can have low volatility and strong autocorrelation without an additional persistent economic factor.

Keywords: term structure models, bond supply, issuance cohort effect, pricing errors.
JEL classification: G12, E43, C38.

* The findings, interpretations, and conclusions expressed in this paper are entirely those of the author and should not be attributed to the Bank of Korea.

† Office of Economic Modeling & Policy Analysis, Bank of Korea, Seoul, South Korea, Email: econ.preference@gmail.com, Tel: +82-2-759-4114.

1 Introduction

A small number of factors explain most movements in the yield curve. As Diebold and Rudebusch (2013) document, the first three principal components (PCs), commonly associated with level, slope, and curvature, account for almost all variation in U.S. Treasury yields. This empirical pattern motivates term-structure models in which a few risk factors determine bond prices across many maturities.

However, estimated term-structure models often leave strongly autocorrelated pricing errors. Adrian, Crump, and Moench (2013) document this pattern even with five pricing factors, while Hamilton and Wu (2014) find serial correlation in measurement or specification errors across the formulations they investigate. Such errors suggest price dynamics that the chosen factors do not capture. Researchers have consequently addressed this persistence through richer model dynamics, more flexible error specifications, or alternative methods of extracting factors. Dempster and Tang (2011) incorporate serially and cross-sectionally correlated measurement errors into estimation. Luo, Pang, and Xu (2021) introduce random level shifts in a dynamic Nelson–Siegel model and report a marked reduction in residual persistence. Li (2025) extracts factors from lagged yield covariances and obtains less autocorrelated residuals than alternative three-factor models.

Figure 1 shows the same pattern of strong autocorrelation in pricing errors documented in earlier studies. It plots the autocorrelations of Treasury yield residuals at lags of 1, 3, 6, and 12 months after the first two PCs are removed.[1] In monthly U.S. Treasury yields at annual maturities from one through thirty years from November 1985 through June 2025,[2] two PCs explain approximately 99.77% of total variance. However, the residuals at individual maturities remain correlated over several months. Thus even a factor approximation with almost no unexplained variance can leave substantial time dependence.

This paper takes a different approach to interpreting residual autocorrelation. I show that the aging of outstanding bonds can generate strong persistence in the residuals without an additional persistent economic factor or a separate error process. I study this mechanism in a preferred-habitat model following Greenwood and Vayanos (2014) and Vayanos and Vila (2021).

Consider an increase in the supply of ten-year bonds today. Risk-averse arbitrageurs who absorb these bonds require compensation for bearing portfolio risk. A year later, the same bonds remain outstanding as nine-year bonds. Since bonds of different maturities share

[1] When maturities below one year are included, three PCs may provide a closer factor approximation. Short-maturity securities also serve as collateral and are held by money market funds subject to maturity mandates (Vayanos and Vila, 2021). Persistent variation in these institutional demands may generate additional residual autocorrelation. Since these features are not separately modeled here, I focus the empirical and quantitative yield comparisons on maturities of at least one year and retain two leading PCs.

[2] This is the paper's main sample; see Section 5.1 for a description of the data.

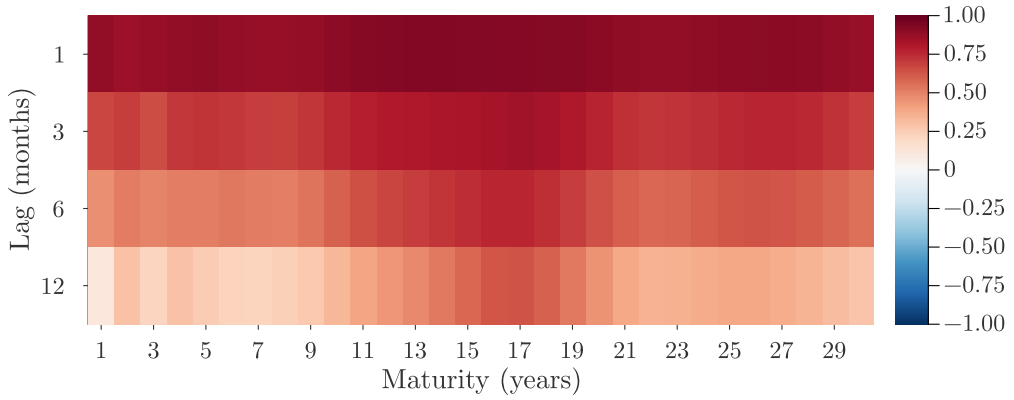


Figure 1: Autocorrelation of observed Treasury yield residuals

Notes: At each annual maturity from one through thirty years, yields are regressed on an intercept and the first two principal components. Colors report residual sample autocorrelations at lags of 1, 3, 6, and 12 months. I use yield data from Filipović, Pelger, and Ye (2026) for November 1985–June 2025.

interest-rate risk, these bonds continue to affect the risk premium on ten-year bonds. Thus, ten-year yields at both dates can depend on the same issuance event, even if new issuance is independent over time. A few factors can capture their broad common price movements while leaving smaller effects in the residuals. These residual price effects can remain strongly autocorrelated because bonds from the same issue continue to affect prices until maturity. I call this continuing influence of bonds from the same issuance on prices the *issuance cohort effect*, or *cohort effect* for short.

I first study the cohort effect in a model with a stochastic short rate and a Poisson process of bond issues with a common maturity M. In this setting, a price approximation for small risk aversion and short-rate volatility gives autocorrelation results for the PCs of the cohort component of log prices. Without short-rate mean reversion and with prices evaluated at a zero short rate, all PCs have the same right derivative of autocorrelation at zero, $\rho_n'(0+) = -2/M$. Here $\rho_n(h)$ denotes the autocorrelation of the nth PC at lag h. Longer issuance maturity brings this lower bound closer to zero, limiting how rapidly autocorrelation can initially decline. For example, with $M = 10$ years, the first-order approximation to one-month autocorrelation is at least 0.983. Thus even higher PCs can remain strongly autocorrelated over short lags. I also show that without short-rate mean reversion, the variance of the nth PC of log prices declines as n^{-8}. Higher PCs therefore account for only a small share of total variance. Thus even PCs with little variance can remain strongly autocorrelated over short lags.

I extend these theoretical results in three directions. First, I allow a positive baseline short rate. Even when the short rate reverts toward a positive long-run mean, all PCs satisfy $-4/M \leq \rho_n'(0+) \leq 0$. Moreover, without short-rate mean reversion, the n^{-8} variance decay also holds at positive baseline short rates. Second, I introduce balance-sheet costs, following He, Nagel, and Song (2022). These costs are proportional to the square of the

total market value of arbitrageurs' bond holdings. At a zero baseline short rate without mean reversion, the initial autocorrelation slopes lie between $-2/M$ and $-1/M$ with positive balance-sheet costs. In addition, the PC variances decline as n^{-4}, so small variances and persistent autocorrelations continue to coexist. Third, since actual bond issuance generally follows a preannounced schedule, I derive the main theoretical results with fixed issuance times. The resulting covariance structure is similar to that under Poisson issuance, and small variances and strong autocorrelations at short lags continue to coexist.

The theoretical analysis highlights four determinants of the size of the cohort effect: issuance maturity, issuance intensity, the second moment of issue size, and balance-sheet costs. Longer issuance maturities keep more cohorts outstanding and lengthen arbitrageurs' exposure to their interest-rate risk. Higher issuance intensity also increases the number of outstanding cohorts, while a larger second moment of issue size increases the variance of supply shocks. Higher balance-sheet costs further amplify the cohort effect. Each cohort occupies arbitrageurs' balance sheets until redemption, so a larger cost raises the compensation required for absorbing the same issuance and magnifies its price effects.

To extend the analysis beyond these approximations and allow issuance maturities to vary, I solve a quantitative variant of the model numerically. The numerical model combines the stochastic short rate and demand factor of Vayanos and Vila (2021) with the cohort effect and a quadratic balance-sheet cost. In the calibrated model, two PCs explain approximately 99.91% of yield variance. The first closely tracks the demand factor, while the second reflects the short-rate factor and the cohort effect. Higher PCs, starting with PC3, reflect the cohort effect but account for only a small share of variance. These PCs all exhibit strong autocorrelation at short lags.

To isolate the contribution of the cohort effect to autocorrelation, I replace future supply by an independent draw from its stationary distribution. This preserves the stationary supply distribution, but future supply no longer follows from the aging of the bonds initially outstanding. The autocorrelation of PC1 is essentially unchanged, while that of PC2 declines and those of higher PCs fall to nearly zero. This confirms that the cohort effect is the main source of strong autocorrelation in higher PCs in the calibrated model.

Strong residual autocorrelation therefore need not indicate that an additional economic factor is missing from a term-structure model. Adding other factors can reduce residual variance while leaving strong autocorrelation if the cohort effect persists. Moreover, a reduction in residual autocorrelation may not by itself identify a missing economic mechanism: the added factors may simply capture the cohort effect indirectly. To assess model misspecification using residual autocorrelation, one should first account for the cohort effect and examine the remaining autocorrelation.

This paper is closely related to the work of Crump and Gospodinov (2022). They show

that smooth dependence across maturities can make a few PCs explain most variation in bond returns even when the underlying factor space has many dimensions. Their result cautions against interpreting a high explained variance as evidence that the omitted variation is economically unimportant. This paper examines the time dependence of that omitted variation. I show that the cohort effect can generate strongly autocorrelated residuals even when a few PCs explain almost all price variance. The cohort effect therefore provides an economic mechanism through which a close factor approximation can leave persistent residuals.

Several papers examine how bond supply affects yields over time. Hayashi (2018) shows that supply shocks can continue to affect yields as outstanding bonds move toward shorter maturities. Costain, Nuño, and Thomas (2025) trace how asset purchases continue to affect yields after purchases end, with these effects fading as the acquired bonds mature. In the model of Barillas and Nimark (2017), supply shocks are independent over time, yet a single shock can continue to affect yields by changing investors' estimates of persistent latent states. The mechanism studied here operates through compensation for holding outstanding bonds. My focus is on how it determines both the size and the persistence of price movements left outside a low-dimensional factor approximation.

2 The model

New issues add to the stock of bonds. Existing bonds move toward shorter remaining maturities and leave the stock when they are repaid. I specify this supply process first, then derive how investors' compensation for holding the bonds affects prices.

2.1 Economic environment

All issues initially share a common maturity, so differences in remaining maturity reflect their issuance dates. This assumption simplifies the analytical derivations. In Section 5, I allow issuance maturities to vary using Treasury auction data. Time and maturity are measured in years. A zero-coupon bond pays one unit of currency at redemption. Issuance times $\{T_i\}$ form a stationary Poisson process on the real line with intensity $\nu > 0$. Every issue has maturity $M > 0$. Its face amount $A_i > 0$ is iid across issues, independent of issuance times, with $\mathbb{E}[A_i^2] < \infty$.

Take stationary mean supply, $\nu\mathbb{E}[A_i]du$, as the reference holdings of preferred-habitat investors at each remaining maturity. Note that their actual holdings are not fixed at these reference levels. This choice simplifies the analysis by centering the supply process.

Outstanding bond supply net of these reference holdings is

$$q_t(du) = \sum_{t-M<T_i\leq t} A_i \delta_{M-t+T_i}(du) - \nu \mathbb{E}[A_i] du, \qquad \text{for } 0 < u \leq M, \tag{2.1}$$

where δ_u is unit mass at maturity u. Gross issuance is positive; a negative centered position means that reference holdings exceed current supply at that maturity.

Write $\langle q, f \rangle = \int_0^M f(u) q(du)$ for supply weighted by a maturity function f. Supply enters the pricing equations through weighted integrals of this form and their conditional expectations. The conditional expectation of weighted supply is

$$\mathbb{E}_t \langle q_{t+s}, f \rangle = \int_{(s,M]} f(u-s) q_t(du), \qquad \text{for } 0 < s < M. \tag{2.2}$$

Equation (2.2) follows by separating future supply into bonds outstanding today and new issues over the next s years. Among current bonds, only those with $u > s$ remain outstanding after s years, when their remaining maturity is $u - s$; this gives the integration range $(s, M]$ and weight $f(u-s)$. For new issues, expected supply cancels the stationary mean supply on $(M-s, M]$, so their net contribution to this conditional expectation is zero. Together, these observations yield Equation (2.2). In addition, taking the right derivative of Equation (2.2) at $s = 0$ gives $\mathcal{L}_q \langle q, f \rangle = -\langle q, f' \rangle$ for differentiable f with $f(0) = 0$, where $\mathcal{L}_q$ is the infinitesimal generator of the supply process.

The instantaneous short rate follows

$$dr_t = \kappa_r(\bar{r} - r_t)dt + \sigma dW_t^r, \qquad \text{with } \kappa_r \geq 0, \quad \bar{r} \geq 0, \quad \text{and} \quad \sigma > 0. \tag{2.3}$$

Here W_t^r is standard Brownian motion, and $\bar{r}$ is the baseline short rate. When $\kappa_r > 0$, it is also the long-run mean of r_t.

Following Vayanos and Vila (2021), preferred-habitat demand depends on a factor β_t satisfying

$$d\beta_t = -\kappa_\beta \beta_t dt + \sigma dW_t^\beta, \qquad \text{with } \kappa_\beta > 0, \tag{2.4}$$

where the demand factor's diffusion coefficient is normalized to σ. In addition, the two Brownian motions and issuance are mutually independent.

Let $P_{a,\lambda}(r, \beta, q, \tau)$ denote a bond price at risk aversion $a \geq 0$, where τ is the bond's remaining maturity and $\lambda \geq 0$ is the balance-sheet cost parameter introduced below. When preferred-habitat investors' demand at maturity u falls below their reference holdings $\nu \mathbb{E}[A_i] du$, arbitrageurs must absorb the difference. Let $D_a(u)$ denote the resulting supply to arbitrageurs, measured by market value per unit of remaining maturity; negative values correspond to preferred-habitat investors' demand above the reference. I specify

$$D_a(u) = \alpha(u) \log P_{a,\lambda}(r, \beta, q, u) + b(u, \beta). \tag{2.5}$$

For $\alpha(u) \geq 0$, preferred-habitat investors' demand, $-D_a(u)$, decreases with the bond price.

The face holdings of preferred-habitat investors relative to their reference holdings are $-D_a(u)du/P_{a,\lambda}(r, \beta, q, u)$. Arbitrageurs' market-clearing face holdings Θ_a are therefore

$$\Theta_a(r, \beta, q; du) = q(du) + \frac{D_a(u)}{P_{a,\lambda}(r, \beta, q, u)} du.$$

Note that subtracting these relative holdings from q cancels the common reference holdings, so the result is total net supply to arbitrageurs.

An arbitrageur with chosen face holdings $\theta_t(du)$ and wealth V_t has a self-financing budget:

$$dV_t = \int \theta_t(du) dP_{a,\lambda}(u) + r_t \left[V_t - \int P_{a,\lambda}(u) \theta_t(du) \right] dt. \tag{2.6}$$

Each price increment is for a held bond, whose remaining maturity falls by dt.

Following He, Nagel, and Song (2022), I add a quadratic balance-sheet cost of aggregate market-value holdings to the instantaneous portfolio-balance objective of Greenwood and Vayanos (2014):

$$\mathbb{E}_t[dV_t] - \frac{a}{2} \operatorname{Var}_t(dV_t) - \frac{\lambda}{2} \left[\int_0^M P_{a,\lambda}(u) \theta_t(du) \right]^2 dt.$$

The last term is the balance-sheet cost. Bond holdings incur balance-sheet costs because leverage constraints and costly external equity financing limit intermediaries' capacity to accommodate additional positions. At market clearing, these holdings are $\int_0^M P_{a,\lambda}(u) q(du) + \int_0^M D_a(u) du$.

Adding a face position at maturity τ, financed from cash, gives the first-order condition:

$$\frac{\mathbb{E}_t[dP_{a,\lambda}(\tau)]}{dt} - r_t P_{a,\lambda}(\tau) = a \frac{\operatorname{Cov}_t(dP_{a,\lambda}(\tau), dV_t)}{dt} + \lambda P_{a,\lambda}(\tau) \int_0^M P_{a,\lambda}(u) \Theta_a(du). \tag{2.7}$$

The right side compensates for the bond's marginal contributions to portfolio variance and balance-sheet costs. Market clearing sets $\theta_t(du) = \Theta_a(r_t, \beta_t, q_t; du)$.

2.2 Bond pricing

I approximate bond prices jointly to first order around $(a, \lambda) = (0, 0)$, where prices equal expected discounted payoffs. To isolate the cohort effect, I first suppress additional demand, $\alpha(u) = b(u, \beta) = 0$, so market clearing gives $\theta_t = q_t$ and prices can be written $P_{a,\lambda}(r, q, \tau)$. I then restore demand and show that the price approximation contains the same cohort component together with a demand component.

At $a = \lambda = 0$, define short-rate duration and the baseline price by

$$B_{\kappa_r}(u) = \begin{cases} (1 - e^{-\kappa_r u})/\kappa_r, & \text{if } \kappa_r > 0, \\ u, & \text{if } \kappa_r = 0 \end{cases}, \tag{2.8}$$

$$\text{and} \quad P_0(r,\tau) = \mathbb{E}_r e^{-\int_0^\tau r_s ds} = \exp\left[-\bar{r}\tau - B_{\kappa_r}(\tau)(r-\bar{r}) + \frac{\sigma^2}{2}\int_0^\tau B_{\kappa_r}(s)^2 ds\right]. \tag{2.9}$$

The accumulated short rate is Gaussian. Its mean is $\bar{r}\tau + B_{\kappa_r}(\tau)(r-\bar{r})$ and its variance is $\sigma^2 \int_0^\tau B_{\kappa_r}(s)^2 ds$, giving Equation (2.9). This is the zero-risk-premium specialization of Vasicek (1977). Since $\partial_r P_0 = -B_{\kappa_r} P_0$, B_{κ_r} is the bond's short-rate duration.

Write $\mathcal{L}_r = \kappa_r(\bar{r} - r)\partial_r + \sigma^2 \partial_{rr}/2$ and $\mathcal{L}_0 = \mathcal{L}_q + \mathcal{L}_r$. The baseline price P_0 in Equation (2.9) satisfies $(\mathcal{L}_0 - \partial_\tau - r)P_0 = 0$. Its held-bond price increment and market-clearing portfolio increment are therefore

$$dP_0(\tau) = rP_0(\tau)dt + \sigma \partial_r P_0(\tau) dW_t^r, \qquad \text{and} \quad dV_t = rV_t dt + \sigma \langle q, \partial_r P_0 \rangle dW_t^r,$$

by Itô's lemma, the market-clearing condition $\theta_t = q_t$, and Equation (2.6). Thus their covariance per unit time is $\sigma^2 \partial_r P_0(r,\tau)\langle q, \partial_r P_0 \rangle$.

To obtain the first-order price approximation around $(a, \lambda) = (0, 0)$, write the formal expansion:

$$P_{a,\lambda} = P_0 + aP_c^{(1)} + \lambda P_\lambda^{(1)} + o(a + \lambda).$$

The subscript c denotes the cohort component of compensation for risk; $P_\lambda^{(1)}$ is the balance-sheet cost coefficient. The superscript denotes expansion order.

The two coefficients can be solved separately at this order. Specifically, matching the coefficient of λ in Equation (2.7) gives

$$\partial_\tau P_\lambda^{(1)} = (\mathcal{L}_0 - r)P_\lambda^{(1)} - P_0(r,\tau)\langle q, P_0(r,\cdot)\rangle, \qquad \text{and} \quad P_\lambda^{(1)}(r,q,0) = 0.$$

Similarly, matching the coefficient of a in Equation (2.7) yields

$$\partial_\tau P_c^{(1)} = (\mathcal{L}_0 - r)P_c^{(1)} - \sigma^2 \partial_r P_0(r,\tau)\langle q, \partial_r P_0 \rangle, \qquad \text{and} \quad P_c^{(1)}(r,q,0) = 0. \tag{2.10}$$

Equation (2.10) defines a linear pricing problem. An admissible solution to this problem is one whose discounted Itô decomposition up to redemption has integrable drift and a true martingale part. Lemma A.1 in Online Appendix A establishes uniqueness in this class and verifies first-order portfolio optimality.

We next evaluate the first-order price coefficient $P_c^{(1)}(r,q,\tau)$ at $r = \bar{r}$ and let short-rate volatility σ tend to zero. The short-rate path then converges to the constant $\bar{r}$, allowing us to

derive analytically how bond prices respond to bond supply.

Proposition 1 (Cohort component of bond prices). *Suppose* $\kappa_r \geq 0$ *and* $\bar{r} \geq 0$. *For* $0 \leq \tau \leq M$, *the admissible solution of Equation* (2.10) *satisfies*

$$\lim_{\sigma \downarrow 0} \frac{-P_c^{(1)}(\bar{r}, q, \tau)}{\sigma^2 P_0(\bar{r}, \tau)} = \langle q, H_{\kappa_r, \bar{r}}(\tau, \cdot) \rangle, \tag{2.11}$$

where

$$H_{\kappa_r, \bar{r}}(\tau, u) = \int_0^{\min(\tau, u)} B_{\kappa_r}(\tau - s) B_{\kappa_r}(u - s) e^{-\bar{r}(u-s)} ds. \tag{2.12}$$

Proof. Discount Equation (2.10) along a bond with remaining maturity $\tau - s$. Integrating to redemption and using its zero terminal coefficient gives

$$P_c^{(1)}(r, q, \tau) = -\sigma^2 \int_0^{\tau} \mathbb{E}_{r,q} \left[e^{-\int_0^s r_\ell d\ell} \partial_r P_0(r_s, \tau - s) \langle q_s, \partial_r P_0(r_s, \cdot) \rangle \right] ds. \tag{2.13}$$

Starting at $r = \bar{r}$, the short-rate path tends to the constant $\bar{r}$ as $\sigma \downarrow 0$. For a remaining maturity ℓ, the baseline price $P_0(r_s, \ell)$ tends to $e^{-\bar{r}\ell}$ and its short-rate derivative to $-B_{\kappa_r}(\ell) e^{-\bar{r}\ell}$. Dividing $e^{-\int_0^s r_\ell d\ell} \partial_r P_0(r_s, \tau - s)$ by $P_0(\bar{r}, \tau)$ cancels its discount factors in this limit, leaving only the discount factor in $\langle q_s, \partial_r P_0(r_s, \cdot) \rangle$:

$$\frac{e^{-\bar{r}s} e^{-\bar{r}(\tau - s)} e^{-\bar{r}(u-s)}}{e^{-\bar{r}\tau}} = e^{-\bar{r}(u-s)}.$$

The uniform integrability established in Lemma A.1 permits taking the limit inside the expectation and integral. Independence of the short rate and issuance, followed by Equation (2.2), gives

$$\begin{aligned} \lim_{\sigma \downarrow 0} \frac{-P_c^{(1)}(\bar{r}, q, \tau)}{\sigma^2 P_0(\bar{r}, \tau)} &= \int_0^{\tau} B_{\kappa_r}(\tau - s) \int_{(s, M]} B_{\kappa_r}(u - s) e^{-\bar{r}(u-s)} q(du) ds \\ &= \int_0^{M} \left[\int_0^{\min(\tau, u)} B_{\kappa_r}(\tau - s) B_{\kappa_r}(u - s) e^{-\bar{r}(u-s)} ds \right] q(du). \end{aligned}$$

The time integral ends when either the target bond or the cohort reaches maturity, whichever comes first. Its upper limit is therefore $\min(\tau, u)$. □

At elapsed time s, the two duration terms, $B_{\kappa_r}(\tau - s)$ and $B_{\kappa_r}(u - s)$, measure the short-rate exposures of the bond being priced and an outstanding issuance cohort with current remaining maturity u, respectively. Their product, multiplied by the cohort's bond price $e^{-\bar{r}(u-s)}$, determines the required compensation at that date. The kernel $H_{\kappa_r, \bar{r}}(\tau, u)$ adds these contributions until the first redemption. Note that current aggregate duration need not

summarize the cohort effect. For example, holdings of shorter- and longer-maturity bonds can be chosen to have the same short-rate exposure today. The shorter-maturity bonds are redeemed sooner, while the longer-maturity bonds continue to expose arbitrageurs to short-rate risk. The two holdings can therefore have different effects on prices today.

In the absence of the demand factor and balance-sheet costs, Proposition 1 gives the first-order approximations in the small-volatility limit at $(r, q) = (\bar{r}, q_t)$:

$$\begin{aligned} \log P_{a,0} - \log P_0 &= a\frac{P_c^{(1)}}{P_0} + o(a) \approx -a\sigma^2\langle q_t, H_{\kappa_r,\bar{r}}(\tau,\cdot)\rangle, \\ \text{and} \quad y_{a,0} - y_0 &= -a\frac{P_c^{(1)}}{\tau P_0} + o(a) \approx \frac{a\sigma^2}{\tau}\langle q_t, H_{\kappa_r,\bar{r}}(\tau,\cdot)\rangle, \end{aligned} \tag{2.14}$$

where $y_{a,0} = -\log P_{a,0}/\tau$ and $y_0 = -\log P_0/\tau$ for $\tau > 0$. Thus, after removing the common factor $-a\sigma^2$, the effect of outstanding supply q_t on log prices across bond maturities is given by the response curve X_t:

$$X_t(\tau) = \langle q_t, H_{\kappa_r,\bar{r}}(\tau,\cdot)\rangle. \tag{2.15}$$

The common scalar $a\sigma^2$ drops out of variance shares and correlations.

Restoring the demand factor while holding $\alpha(u)$ and $b(u,\beta)$ fixed gives the first-order decomposition:

$$\log P_{a,\lambda}(r,\beta,q,\tau) = \log P_0(r,\tau) + a\frac{P_d^{(1)}(r,\beta,\tau)}{P_0(r,\tau)} + a\frac{P_c^{(1)}(r,q,\tau)}{P_0(r,\tau)} + \lambda\frac{P_\lambda^{(1)}(r,\beta,q,\tau)}{P_0(r,\tau)} + o(a+\lambda),$$

where $P_d^{(1)}$ is the demand component of the price coefficient on a. The balance-sheet cost coefficient $P_\lambda^{(1)}(r,\beta,q,\tau)$ also separates into demand and cohort components. Online Appendix A derives this full decomposition of log bond prices:

$$\begin{aligned} \log P_{a,\lambda}(r,\beta,q,\tau) = \log P_0(r,\tau) &+ a\frac{P_d^{(1)}(r,\beta,\tau)}{P_0(r,\tau)} + a\frac{P_c^{(1)}(r,q,\tau)}{P_0(r,\tau)} \\ &+ \lambda\frac{P_{\lambda,d}^{(1)}(r,\beta,\tau)}{P_0(r,\tau)} + \lambda\frac{P_{\lambda,c}^{(1)}(r,q,\tau)}{P_0(r,\tau)} + o(a+\lambda), \end{aligned} \tag{2.16}$$

where $P_{\lambda,d}^{(1)}$ and $P_{\lambda,c}^{(1)}$ are the demand and cohort components of $P_\lambda^{(1)}$, respectively. At first order, the demand and cohort components have zero cross-covariance, including their balance-sheet cost contributions.[3]

Figure 2 illustrates the pricing formula in Equation (2.14). It shows how one additional unit

[3]Because $P_c^{(1)}$ is linear in centered supply, independence gives $\mathbb{E}[P_c^{(1)}/P_0 \mid r_t, \beta_t] = 0$. Since $P_d^{(1)}/P_0$ depends only on (r_t, β_t), $\mathbb{E}[P_d^{(1)}P_c^{(1)}/P_0^2] = 0$. The same argument applies to the balance-sheet cost components $P_{\lambda,d}^{(1)}$ and $P_{\lambda,c}^{(1)}$.

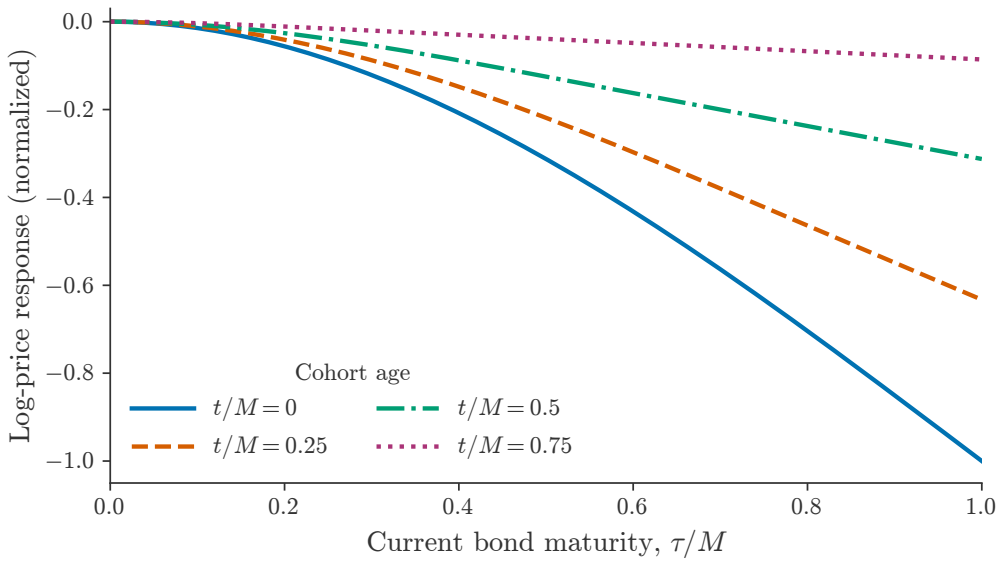


Figure 2: Effects of additional bond supply on log prices

Notes: Each line shows how one additional unit issued at maturity M affects log prices t years after issuance. The added bond has remaining maturity $M - t$; τ is the current remaining maturity of the bond being priced. Here $\bar{r} = \kappa_r = 0$, and all curves are normalized by the same factor so the change in log price at $t = 0$ and $\tau = M$ is -1.

issued at maturity M affects log prices as it ages, with a zero baseline short rate and $\kappa_r = 0$. The added unit raises $X_t(\tau)$ by $H_{0,0}(\tau, M-t)$ and changes the log price by $-a\sigma^2 H_{0,0}(\tau, M-t)$. Accordingly, it plots $-H_{0,0}(\tau, M-t)/H_{0,0}(M, M)$, normalized so the response at $t = 0$ and $\tau = M$ is -1. This additional supply can also be thought of as an exogenous sale of one unit at remaining maturity $M - t$ by preferred-habitat investors, provided their holdings of that bond remain one unit lower until redemption. Each line fixes the elapsed time t since issuance: the added bond now has remaining maturity $M - t$, while the horizontal axis varies the current remaining maturity τ of the bond being priced. For example, the point $\tau/M = 0.4$ on the $t/M = 0.25$ line shows how the added bond, now with remaining maturity $0.75M$, affects the price of a bond with remaining maturity $0.4M$. In this figure, a longer remaining maturity of either the added bond or the bond being priced increases its short-rate duration, raising $H_{0,0}(\tau, M-t)$ in Equation (2.12). Equation (2.11) then implies a larger decline in log price. The decline shrinks to zero as the added issue approaches maturity.

3 Theoretical results

In the limit $\lambda \to 0$, the response curve X_t describes the cohort component of log bond prices under the approximations in Equation (2.14). This section sets $\bar{r} = 0$ to derive the variances and autocorrelations of its principal components. I first calculate the covariances across maturities and dates. I then derive a lower bound on the right derivative of autocorrelation at zero for $\kappa_r \geq 0$. As issuance maturity M increases, the lower bound approaches zero,

implying strong autocorrelation over sufficiently short lags for each fixed principal component. Finally, for $\kappa_r = 0$, I derive the variances of the principal components and show that higher components account for very little of the total variance.

Section 4 derives results under partially relaxed restrictions. Specifically, Section 4.1 extends these results to $\bar{r} \geq 0$ while retaining their respective assumptions on κ_r. Section 4.2 considers the combined cohort effects of interest-rate risk and balance-sheet costs at $\bar{r} = \kappa_r = 0$. Section 4.3 extends the analysis to deterministic issuance times.

3.1 Covariance structure

The compensated-Poisson covariance identity (Last and Penrose, 2017) gives, for deterministic square-integrable weights,

$$\mathrm{Cov}(\langle q_t, f\rangle, \langle q_t, g\rangle) = \nu\mathbb{E}[A_i^2]\int_0^M f(u)g(u)du. \tag{3.1}$$

At dates t and $t+h$, only shared cohorts contribute to covariance. A cohort with remaining maturity $u+h$ at t has maturity u at $t+h$, so

$$\mathrm{Cov}(\langle q_t, f\rangle, \langle q_{t+h}, g\rangle) = \nu\mathbb{E}[A_i^2]\int_0^{M-h} f(u+h)g(u)du, \qquad \text{for } 0 \leq h < M. \tag{3.2}$$

At $h \geq M$, the issuance windows are disjoint and this covariance is zero.

For square-integrable functions f and g on the maturity interval $[0, M]$, we define the inner product $\langle f, g\rangle = \int_0^M f(u)g(u)du$. We also define the integral operator:

$$(\mathcal{G}_{\kappa_r} f)(\tau) = \int_0^M H_{\kappa_r,0}(\tau, u)f(u)du.$$

Since $H_{\kappa_r,0}$ is symmetric by Equation (2.12), $\langle f, X_t\rangle = \langle q_t, \mathcal{G}_{\kappa_r} f\rangle$ holds. Applying this identity to Equation (3.1) identifies the covariance operator of X_t:

$$\mathrm{Cov}(\langle f, X_t\rangle, \langle g, X_t\rangle) = \nu\mathbb{E}[A_i^2]\langle \mathcal{G}_{\kappa_r} f, \mathcal{G}_{\kappa_r} g\rangle = \langle f, \nu\mathbb{E}[A_i^2]\mathcal{G}_{\kappa_r}^2 g\rangle. \tag{3.3}$$

Let ϕ_n be orthonormal eigenfunctions of $\mathcal{G}_{\kappa_r}$, ordered by nonincreasing eigenvalues $\eta_n > 0$. Lemma A.2 shows that $\mathcal{G}_{\kappa_r}$ satisfies the conditions for this eigenfunction decomposition. By Equation (3.3), the covariance operator of X_t is $\nu\mathbb{E}[A_i^2]\mathcal{G}_{\kappa_r}^2$, so ϕ_n are also the principal component directions of X_t. The score of component n and its variance are

$$F_{n,t} = \langle \phi_n, X_t\rangle = \eta_n\langle q_t, \phi_n\rangle, \qquad \text{and} \quad \mathrm{Var}(F_{n,t}) = \nu\mathbb{E}[A_i^2]\eta_n^2. \tag{3.4}$$

Different scores are uncorrelated at the same date.

The autocorrelations of the principal components $F_{n,t}$ follow from Equation (3.2):

$$\rho_n(h) = \mathrm{Corr}(F_{n,t}, F_{n,t+h}) = \int_0^{M-h} \phi_n(u+h)\phi_n(u)du, \qquad \text{for } 0 \le h < M. \tag{3.5}$$

The eigenvalue η_n scales a component's variance but cancels from its autocorrelation. Correlation instead depends on the shape of ϕ_n, which also changes with rank. To compare persistence across components, we need the restrictions that the pricing equation places on these eigenfunctions.

3.2 Persistence of the principal components

By Equation (3.4), a unit of new issuance at maturity M changes the score by $\eta_n\phi_n(M)$. New issues are independent of current supply, so those arriving between t and $t+h$ add variation to $F_{n,t+h}$ that is not shared with $F_{n,t}$. Relative to the variance of the component, this contribution is proportional to $\phi_n(M)^2$ and lowers autocorrelation at short lags. The following theorem establishes this result formally under $\bar{r} = 0$. Section 4.1 extends the result to $\bar{r} \ge 0$ for $\kappa_r \ge 0$.

Theorem 1 (Persistence of principal components at short lags). *For $\kappa_r \ge 0$, $\bar{r} = 0$, and each n,*

$$\rho_n'(0+) = -\frac{\phi_n(M)^2}{2} = -\frac{2}{M} + \frac{\eta_n}{2M}\left[\kappa_r\phi_n'(M)^2 + 2\kappa_r^2\int_0^M \phi_n'(u)^2du\right] \in \left[-\frac{2}{M}, 0\right]. \tag{3.6}$$

Proof. The eigenfunctions are smooth and satisfy $\phi_n(0) = 0$. Differentiating Equation (3.5) from the right therefore gives

$$\rho_n'(0+) = -\phi_n(M)^2 + \int_0^M \phi_n\phi_n'du = -\frac{\phi_n(M)^2}{2}. \tag{3.7}$$

To evaluate the endpoint, suppress n and put

$$N = \int_0^M \phi^2du, \quad U_1 = \int_0^M (\phi'')^2du, \quad U_0 = \int_0^M (\phi')^2du, \quad \text{and} \quad U_\partial = \phi'(M)^2.$$

Lemma A.2 supplies the differential equation and boundary conditions:

$$\phi'''' - \kappa_r^2\phi'' = \frac{\phi}{\eta}, \tag{3.8}$$

$$\phi(0) = \phi'(0) = 0, \quad \phi''(M) = -\kappa_r\phi'(M), \quad \text{and} \quad \phi'''(M) = \kappa_r^2\phi'(M).$$

Multiplying the differential equation by ϕ and integrating by parts twice yields

$$\begin{aligned}\frac{N}{\eta} &= [\phi\phi''' - \phi'\phi'' - \kappa_r^2\phi\phi']_0^M + U_1 + \kappa_r^2 U_0 \\ &= U_1 + \kappa_r U_\partial + \kappa_r^2 U_0. \end{aligned} \tag{3.9}$$

Multiplying the right side of the differential equation in Equation (3.8) by $u\phi'$ and denoting the resulting integral by I gives

$$I = \frac{1}{\eta}\int_0^M u\phi'\phi\, du = \frac{1}{2\eta}\{M\phi(M)^2 - N\}. \tag{3.10}$$

Multiplying each term on the left side of Equation (3.8) by $u\phi'$ and integrating by parts using the boundary conditions gives

$$\int_0^M u\phi'\phi''''du = [u\phi'\phi''' - \phi'\phi'' - u(\phi'')^2/2]_0^M + \frac{3}{2}U_1 = \frac{3}{2}U_1 + \kappa_r U_\partial + \frac{M\kappa_r^2 U_\partial}{2},$$

and $$-\kappa_r^2\int_0^M u\phi'\phi''du = -\frac{M\kappa_r^2 U_\partial}{2} + \frac{\kappa_r^2 U_0}{2}.$$

The sum of these two expressions is I. Hence,

$$I = \frac{3}{2}U_1 + \kappa_r U_\partial + \frac{1}{2}\kappa_r^2 U_0 = \frac{3N}{2\eta} - \frac{1}{2}\kappa_r U_\partial - \kappa_r^2 U_0.$$

The last equality follows by solving Equation (3.9) for U_1 and substituting the result. Comparison with Equation (3.10) yields

$$M\phi(M)^2 = 4N - \eta(\kappa_r U_\partial + 2\kappa_r^2 U_0).$$

Setting $N = 1$ and substituting this expression into Equation (3.7) yields Equation (3.6). □

Equation (3.6) shows that without mean reversion ($\kappa_r = 0$), the right derivative of autocorrelation at zero is $-2/M$ for every component. With mean reversion ($\kappa_r > 0$), these derivatives can differ but remain in $[-2/M, 0]$. Thus higher-order principal components need not have more negative derivatives than lower-order principal components. Longer issuance maturity brings the lower bound closer to zero for each fixed component: bonds remain outstanding longer, allowing the effects of past issues on log prices to persist in the residuals. With maturities and lags measured in years, the one-month first-order approximation implied by Equation (3.6) is at least $1 - 2/(12M)$, approximately 0.983 for $M = 10$ and 0.994 for $M = 30$.

3.3 Variances of the principal components

By Equation (3.4), the variance of component n is $\nu\mathbb{E}[A_i^2]\eta_n^2$. For $\kappa_r > 0$, the eigenvalues are determined by Equation (3.8) and the boundary conditions. At $\kappa_r = 0$, a simpler eigenvalue formula is available. In this case, $B_0(u) = u$, so Equation (2.12) at $\bar{r} = 0$ gives

$$H_{0,0}(\tau, u) = \frac{\max(\tau, u)\min(\tau, u)^2}{2} - \frac{\min(\tau, u)^3}{6}. \tag{3.11}$$

On the unit interval, this is the covariance kernel of once-integrated Brownian motion. Its eigenvalues are given by Freedman (1999). This result yields the following theorem. Section 4.1 extends the variance result to $\bar{r} \geq 0$ while retaining $\kappa_r = 0$.

Theorem 2 (Variances of the principal components). *Suppose $\kappa_r = 0$ and $\bar{r} = 0$. Let ζ_n be the nth positive root of $\cos\zeta\cosh\zeta = -1$. Then,*

$$\begin{aligned} \zeta_n &\sim (n - \tfrac{1}{2})\pi, \qquad \eta_n = \frac{M^4}{\zeta_n^4}, \\ \textit{and} \quad \mathrm{Var}(F_{n,t}) &= \frac{\nu\mathbb{E}[A_i^2]M^8}{\zeta_n^8} \sim \frac{\nu\mathbb{E}[A_i^2]M^8}{[(n - \frac{1}{2})\pi]^8}, \qquad \textit{as } n \to \infty. \end{aligned} \tag{3.12}$$

Proof. Freedman's (1999) unit-interval eigenvalues are ζ_n^{-4}. Substitute $\tau = Mx$ and $u = My$ in the eigenvalue equation $\int_0^M H_{0,0}(\tau, u)\phi_n(u)du = \eta_n\phi_n(\tau)$ for the kernel in Equation (3.11). Since $H_{0,0}(Mx, My) = M^3 H_{0,0}(x, y)$ and $du = Mdy$, the operator eigenvalues on $(0, M)$ are $M^4\zeta_n^{-4}$. Equation (3.4) then gives the variance in Equation (3.12).

The root equation is $\cos\zeta = -1/\cosh\zeta$. Its large positive roots approach the successive positive zeros of cosine, so $\zeta_n \sim (n - \frac{1}{2})\pi$ (Freedman, 1999). □

Theorem 2 shows how the size of the cohort effect in the PCs depends on issuance: every PC's variance increases with issuance maturity M, issuance intensity ν, and the second moment of issue size $\mathbb{E}[A_i^2]$. Longer issuance maturities keep more cohorts outstanding and lengthen arbitrageurs' exposure to their interest-rate risk. Higher issuance intensity also increases the number of outstanding cohorts, whereas a larger second moment of issue size increases the variance of supply shocks. Because the same outstanding cohorts affect bond prices across time and maturities, these changes increase the variation in prices generated by the cohort effect.

After retaining N_{PC} principal components, the omitted curve is $\sum_{n>N_{\mathrm{PC}}} F_{n,t}\phi_n$. Orthonormality and Equation (3.12) give its total variance:[4]

[4]The last asymptotic relation follows from Equation (3.12) and $\sum_{n>N_{\mathrm{PC}}} n^{-8} \sim \int_{N_{\mathrm{PC}}}^{\infty} x^{-8}dx = N_{\mathrm{PC}}^{-7}/7$.

$$\mathbb{E}\left\|\sum_{n>N_{\mathrm{PC}}} F_{n,t}\phi_n\right\|_{L^2}^2 = \sum_{n>N_{\mathrm{PC}}} \mathrm{Var}(F_{n,t}) \sim \frac{\nu\mathbb{E}[A_i^2]M^8}{7\pi^8} N_{\mathrm{PC}}^{-7}, \qquad \text{as } N_{\mathrm{PC}} \to \infty.$$

Together with the common right derivative $\rho_n'(0+) = -2/M$ in Theorem 1, this shows that the remaining variation decreases rapidly as N_{PC} increases, while the initial autocorrelation slopes of the omitted components do not depend on N_{PC}. Thus, improving fit by retaining more PCs need not eliminate strong residual autocorrelation at short lags. Moreover, increasing M raises residual variance and slows the initial decline in autocorrelation. As discussed above, longer maturities keep more cohorts outstanding and allow them to affect bond prices for longer.

3.4 Illustrations of the main results

I conclude this section with graphical illustrations of the main results of Theorems 1 and 2. Figure 3 shows the variance of the nth principal component. The asymptotic result that variances decline as n^{-8} closely approximates the variances of higher components. Thus, residual variation due to the cohort effect declines rapidly as the dimension of the factor approximation increases. Figure 4 plots the corresponding autocorrelations computed from Equation (3.5). All autocorrelation curves in panel (a) have right derivative -2 at zero with respect to h/M, while their correlations differ at positive lags. PC3 has only 1.05×10^{-5} of PC1's variance, but its autocorrelation at $h/M = 0.01$ is 0.978. These numerical calculations

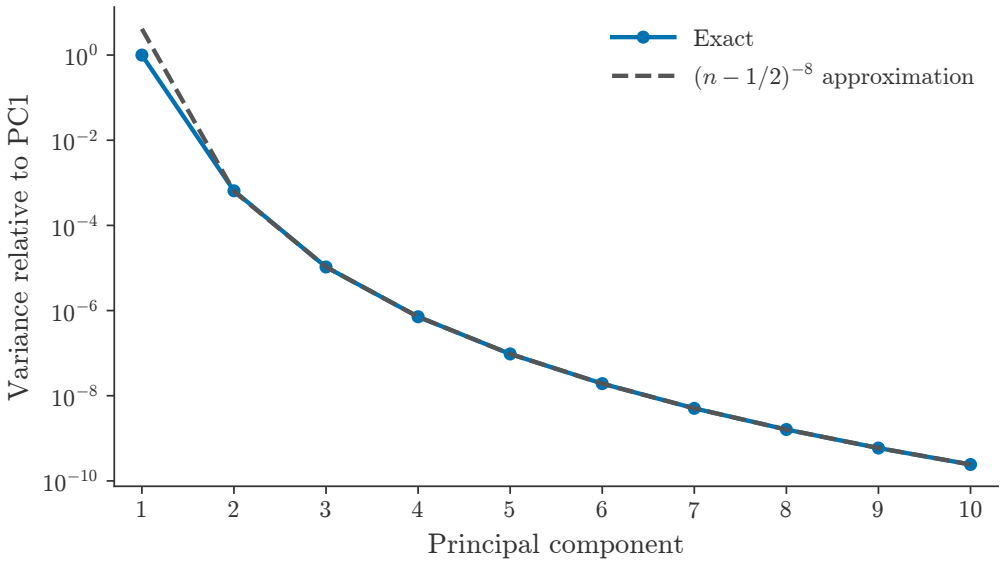


Figure 3: Variances of the principal components with a zero baseline short rate and no mean reversion

Notes: Exact variances are divided by PC1's variance, giving $(\zeta_1/\zeta_n)^8$. The dashed line substitutes $(n - \frac{1}{2})\pi$ for ζ_n, retaining the PC1 normalization. The vertical axis is logarithmic.

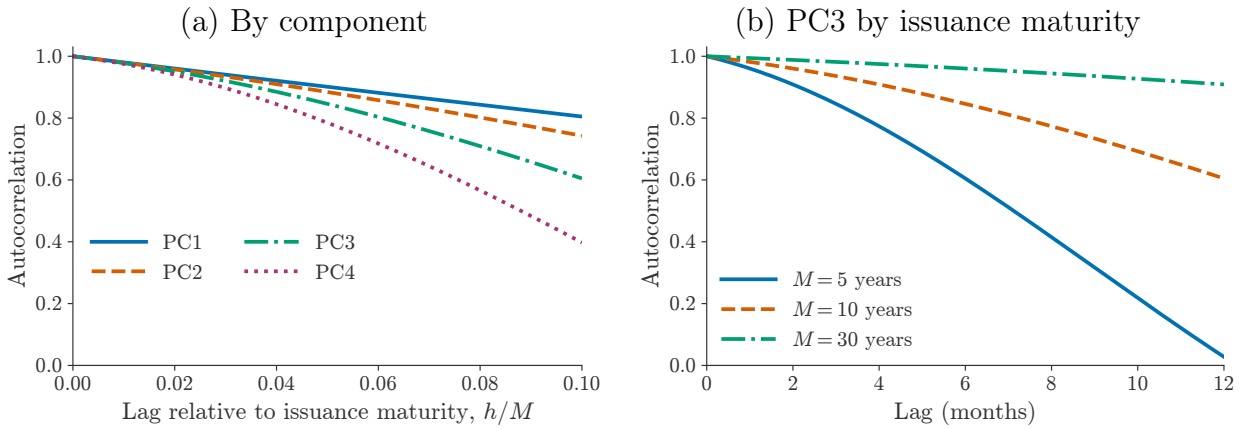


Figure 4: Factor autocorrelations with a zero baseline short rate and no mean reversion

Notes: Curves are computed from Equation (3.5). Panel (a) shows autocorrelations for PC1, PC2, PC3, and PC4, with lags relative to the common issuance maturity M. Panel (b) shows PC3 autocorrelations for $M = 5, 10, 30$ years, with lags measured in months.

confirm that components with very little variance can remain highly correlated over short lags.

Panel (b) of Figure 4 compares the autocorrelations of PC3 at issuance maturities of 5, 10, and 30 years. The one-month autocorrelations are 0.960, 0.982, and 0.994, respectively. At twelve months, they are 0.028, 0.605, and 0.909. A shorter issuance maturity M brings redemption forward, so arbitrageurs bear the interest-rate risk of each cohort for less time. The cohort effect therefore influences prices over a shorter horizon, and autocorrelations decline more rapidly. Conversely, a longer M keeps more past issues outstanding at a fixed issuance intensity, allowing the effects of more independent cohorts to accumulate in prices and amplifying them through longer exposure to interest-rate risk.

4 Extensions

I now return to the general baseline short rate in Section 2 and show that components can have small variances and persistent autocorrelations at positive baseline short rates. I also examine how balance-sheet costs affect these results and whether the main conclusions continue to hold with deterministic issuance times.

4.1 Positive baseline short rates

A positive baseline short rate changes the value and short-rate exposure of outstanding bonds through the discount factor in Equation (2.12). The kernel becomes asymmetric when $\bar{r} > 0$, and discounting changes the covariance operator of the response curve X_t. The principal components therefore also change. The following two results retain the common issuance

maturity M and extend Theorems 1 and 2 separately, preserving their respective assumptions on κ_r.

Write $\mathcal{G} = \mathcal{G}_{\kappa_r,\bar{r}}$ for the integral operator on $L^2(0, M)$ defined by

$$(\mathcal{G}f)(\tau) = \int_0^M H_{\kappa_r,\bar{r}}(\tau, u)f(u)du.$$

Let $\mathcal{G}^*$ denote its adjoint. Equation (2.15) and an interchange of integrals yield $\langle f, X_t\rangle = \langle q_t, \mathcal{G}^* f\rangle$, so Equation (3.1) gives

$$\begin{aligned}\mathrm{Cov}(\langle f, X_t\rangle, \langle g, X_t\rangle) &= \nu\mathbb{E}[A_i^2]\langle \mathcal{G}^* f, \mathcal{G}^* g\rangle \\ &= \langle f, \nu\mathbb{E}[A_i^2]\mathcal{G}\mathcal{G}^* g\rangle.\end{aligned} \tag{4.1}$$

Thus the covariance operator of X_t is $\nu\mathbb{E}[A_i^2]\mathcal{G}\mathcal{G}^*$. Choose real orthonormal left and right singular functions, ordered by $s_1 \geq s_2 \geq \cdots > 0$, satisfying

$$\mathcal{G}\psi_n = s_n\phi_n, \quad \mathcal{G}^*\phi_n = s_n\psi_n, \quad \text{and} \quad \|\phi_n\| = \|\psi_n\| = 1. \tag{4.2}$$

Equation (4.2) implies

$$\nu\mathbb{E}[A_i^2]\mathcal{G}\mathcal{G}^*\phi_n = \nu\mathbb{E}[A_i^2]\mathcal{G}(s_n\psi_n) = \nu\mathbb{E}[A_i^2]s_n^2\phi_n.$$

Thus ϕ_n is an eigenfunction of the price covariance operator, with eigenvalue $\nu\mathbb{E}[A_i^2]s_n^2$, and hence a price principal component direction. Its score satisfies

$$F_{n,t} = \langle \phi_n, X_t\rangle = \langle q_t, \mathcal{G}^*\phi_n\rangle = s_n\langle q_t, \psi_n\rangle, \tag{4.3}$$

so ψ_n weights outstanding supply across remaining maturities to determine that score. At $\bar{r} = 0$, they coincide and $s_n = \eta_n$. As shown in Online Appendix B, the variance is $\nu\mathbb{E}[A_i^2]s_n^2$, while the autocorrelation is $\int_0^{M-h} \psi_n(u + h)\psi_n(u)du$ and does not depend on s_n.

Proposition 2 (Persistence with discounting). *Let $F_{n,t} = \langle \phi_n, X_t\rangle$ denote the principal component scores, and let ρ_n denote their autocorrelations. For $\bar{r} \geq 0$, $\kappa_r \geq 0$, and every fixed n,*

$$-\frac{4}{M} \leq \rho_n'(0+) = d_n \leq 0, \qquad \textit{where} \quad d_n := -\frac{\psi_n(M)^2}{2}, \tag{4.4}$$

and

$$\rho_n(h) = 1 + d_n h + O((h/M)^2), \qquad \textit{as } h \downarrow 0. \tag{4.5}$$

Proof. See Online Appendix B. □

At $\bar{r} = 0$, the lower bound on $\rho_n'(0+)$ sharpens to $-2/M$, with equality when also

$\kappa_r = 0$. Thus issuance maturity continues to provide a lower bound on the right derivative of autocorrelation at zero even with both mean reversion and discounting. With maturities and lags measured in years, the one-month first-order approximation satisfies $1 + d_n/12 \geq 1 - 4/(12M)$, giving lower bounds of approximately 0.967 for $M = 10$ and 0.989 for $M = 30$.

The following proposition extends Theorem 2 to $\bar{r} > 0$. When $\kappa_r = 0$, the short rate follows a Brownian motion with volatility σ. Prices are evaluated conditional on $r = \bar{r}$ at the pricing date, so the short-rate path starts from $\bar{r}$ and converges to the constant $\bar{r}$ as $\sigma \to 0$. Here $\bar{r}$ serves as the current short-rate level at which prices are evaluated, rather than a long-run mean.

Proposition 3 (Variances of the principal components with discounting). *Suppose $\kappa_r = 0$ and $\bar{r} \geq 0$. Define $\zeta_n > 0$ by $\zeta_n^2[\zeta_n^2 + (\bar{r}M)^2] = M^4/s_n$, ordered increasingly and counting multiplicity. Then,*

$$\mathrm{Var}(F_{n,t}) = \frac{\nu\mathbb{E}[A_i^2]M^8}{\zeta_n^4[\zeta_n^2 + (\bar{r}M)^2]^2} \sim \frac{\nu\mathbb{E}[A_i^2]M^8}{(\pi n)^8}, \qquad \textit{and} \quad \zeta_n \sim \pi n. \tag{4.6}$$

The asymptotics hold as $n \to \infty$ for fixed M and $\bar{r}$.

Proof. See Online Appendix C, which also gives the differential equation and boundary conditions that determine ζ_n. □

At $\bar{r} = 0$, ζ_n reduces to the roots of $\cos\zeta\cosh\zeta = -1$, recovering Theorem 2. With positive discounting, the variances of the principal components still decline rapidly. Proposition 2 gives $-4/M \leq \rho_n'(0+) \leq 0$, with or without mean reversion. Thus components with little variance can also remain serially correlated at positive baseline short rates.

4.2 Cohort effects with balance-sheet costs

I now retain both $aP_c^{(1)}/P_0$ and $\lambda P_{\lambda,c}^{(1)}/P_0$ in the first-order log-price approximation. Suppose $\kappa_r = \bar{r} = 0$ and evaluate prices at $r = 0$. The cohort effect in Equation (2.16) has the following small-volatility approximation:

$$a\frac{P_c^{(1)}}{P_0} + \lambda\frac{P_{\lambda,c}^{(1)}}{P_0} \approx -\widetilde{X}_t(\tau).$$

Online Appendix D shows that

$$\widetilde{X}_t(\tau) = \langle q_t, \widetilde{H}_{a,\lambda}(\tau,\cdot)\rangle, \qquad \text{where} \quad \widetilde{H}_{a,\lambda}(\tau,u) = a\sigma^2 H_{0,0}(\tau,u) + \lambda\min(\tau,u). \tag{4.7}$$

This section derives the variances and persistence of the principal components of $\widetilde{X}_t$.

Let $\widetilde{\mathcal{G}}_{a,\lambda}$ be the integral operator $(\widetilde{\mathcal{G}}_{a,\lambda}f)(\tau) = \int_0^M \widetilde{H}_{a,\lambda}(\tau,u)f(u)du$, and let $\widetilde{\phi}_n$ be its orthonormal eigenfunctions, ordered by nonincreasing eigenvalues $\widetilde{\eta}_n > 0$. For fixed $a\sigma^2 \geq 0$ and $\lambda \geq 0$, not both zero, this operator is compact, self-adjoint, and strictly positive (see Online Appendix D). Applying Equations (3.3)–(3.5) to this operator gives the PC scores, variances, and autocorrelations of $\widetilde{X}_t$:

$$\begin{aligned} \widetilde{F}_{n,t} = \widetilde{\eta}_n\langle q_t, \widetilde{\phi}_n\rangle, \qquad \mathrm{Var}(\widetilde{F}_{n,t}) = \nu\mathbb{E}[A_i^2]\widetilde{\eta}_n^2, \\ \text{and} \quad \widetilde{\rho}_n(h) = \int_0^{M-h} \widetilde{\phi}_n(u+h)\widetilde{\phi}_n(u)du, \qquad \text{for } 0 \leq h < M. \end{aligned} \tag{4.8}$$

Based on Equation (4.8), the following proposition characterizes the initial decline in the autocorrelation of $\widetilde{F}_{n,t}$.

Proposition 4 (Persistence with balance-sheet costs). *For the combined kernel in Equation (4.7), define*

$$\omega_n = \frac{a\sigma^2\langle\widetilde{\phi}_n, \mathcal{G}_0\widetilde{\phi}_n\rangle}{\widetilde{\eta}_n} \in [0,1].$$

Then, for every n,

$$\widetilde{\rho}_n'(0+) = -\frac{\widetilde{\phi}_n(M)^2}{2} = -\frac{1+\omega_n}{M} \in \left[-\frac{2}{M}, -\frac{1}{M}\right]. \tag{4.9}$$

The slope equals $-2/M$ when $\lambda = 0$ and $-1/M$ when $a = 0$.

Proof. See Online Appendix D. □

Thus the initial autocorrelation slope remains within $2/M$ of zero independently of component rank, even with positive balance-sheet costs. Autocorrelation therefore remains high at short lags. For example, with $M = 10$ years, the first-order approximation to one-month autocorrelation lies between 0.983 and 0.992. The next result shows how balance-sheet costs change the rate at which PC variances decline.

Proposition 5 (Variances of the principal components with balance-sheet costs). *For fixed $a\sigma^2 \geq 0$, $\lambda > 0$, and M, the combined kernel in Equation (4.7) gives*

$$\widetilde{\eta}_n \sim \frac{\lambda M^2}{[(n-\frac{1}{2})\pi]^2}, \qquad \text{and} \quad \mathrm{Var}(\widetilde{F}_{n,t}) \sim \frac{\nu\mathbb{E}[A_i^2]\lambda^2 M^4}{[(n-\frac{1}{2})\pi]^4}, \qquad \text{as } n \to \infty. \tag{4.10}$$

Proof. See Online Appendix D. □

Proposition 5 describes the effect of balance-sheet costs on PC variances: increasing λ raises every PC's variance because the additional kernel $\min(\tau, u)$ defines a positive operator.

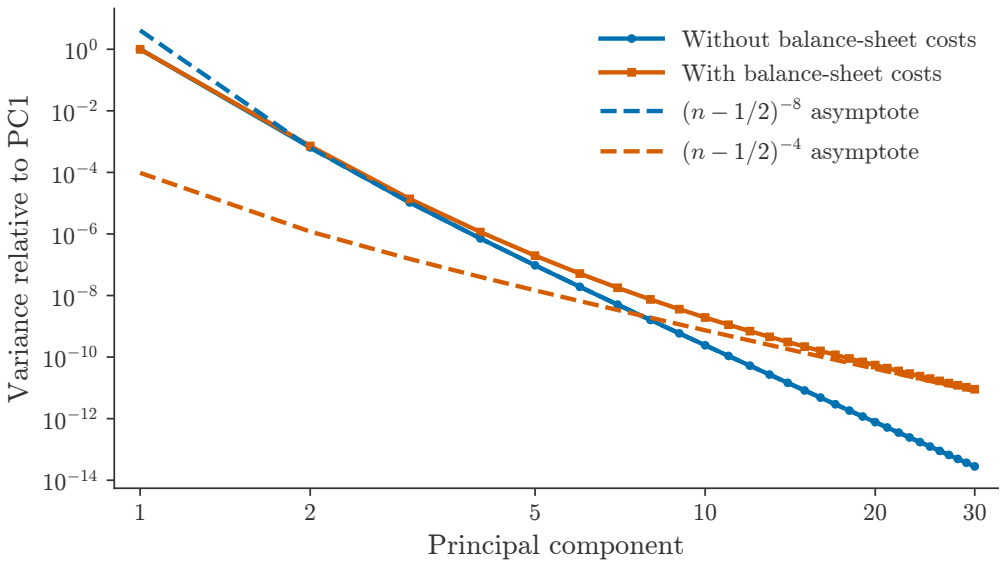


Figure 5: Variances of the principal components with and without balance-sheet costs

Notes: With $\kappa_r = \bar{r} = 0$ in Equation (4.7), the blue and red curves correspond to $\lambda/(a\sigma^2 M^2) = 0$ and 0.00198, respectively. Variances are divided by PC1's variance in the same case. Dashed lines show the corresponding asymptotic expressions in Theorem 2 and Proposition 5. Both axes are logarithmic.

It further shows that positive balance-sheet costs slow the asymptotic decay of PC variances from n^{-8} to n^{-4}, increasing the relative importance of higher components. Nevertheless, the variances of higher components still decay rapidly, while their autocorrelations remain high at short lags, as implied by Equation (4.9).

Figure 5 compares PC variances with and without balance-sheet costs. Introducing balance-sheet costs increases the variances of higher PCs by a much larger proportion than those of lower PCs. In the illustrated case with $\lambda/(a\sigma^2 M^2) = 0.00198$[5], the variances of lower PCs decline faster than the n^{-4} asymptotic rate, while those of higher PCs approach that rate. Thus, introducing balance-sheet costs increases the variances of higher PCs relative to those of lower PCs. As the calibrated model in Section 5 shows, higher PCs of yields primarily reflect the cohort effect. Balance-sheet costs therefore provide a mechanism for amplifying the cohort effect. This amplification arises because each cohort occupies arbitrageurs' balance sheets until redemption: higher costs raise the compensation required to absorb the same issuance and magnify its price effects.

[5]Rescaling maturities to $[0,1]$ gives

$$\widetilde{H}_{a,\lambda}(\tau,u) = a\sigma^2 M^3 \left[H_{0,0}\Big(\frac{\tau}{M},\frac{u}{M}\Big) + \frac{\lambda}{a\sigma^2 M^2} \min\Big(\frac{\tau}{M},\frac{u}{M}\Big) \right].$$

Including the change in the integration variable, the eigenvalues $\widetilde{\eta}_n$ of $\widetilde{\mathcal{G}}_{a,\lambda}$ share the common multiplier $a\sigma^2 M^4$. This factor cancels in eigenvalue ratios, so the ratios of PC variances can be computed using the kernel in brackets instead. I set $\lambda/(a\sigma^2 M^2) = 0.00198$ using the calibrated a, σ, and λ and the maximum issuance maturity M in Section 5.

4.3 Deterministic issuance times

The Poisson issuance process facilitates the preceding derivations, whereas actual bond issuance often follows a preannounced schedule. This section fixes issuance times and shows that the main conclusions of Theorems 1 and 2 continue to hold.

Suppose bonds of maturity M are issued at $T_i = i\Delta$ for $i \in \mathbb{Z}$, with a fixed interval $\Delta > 0$. Issue sizes A_i remain independent and identically distributed, independent of the short rate, and are revealed at issuance, with $0 < \mathrm{Var}(A_i) < \infty$. Independence across issues allows me to show that principal components can exhibit strong autocorrelation even without persistence in issue sizes A_i. I also retain $\bar{r} = \lambda = 0$ and the first-order, small-volatility approximation in Section 3. Expected outstanding supply now depends on the position within the issuance cycle. Subtracting this date-specific mean gives

$$\widehat{q}_t(du) = \sum_{t-M<T_i\leq t} (A_i - \mathbb{E}[A_i])\delta_{M-t+T_i}(du).$$

This centering removes the predictable calendar component of supply and its corresponding mean price response. Future centered issues have zero conditional mean, so the aging relation in Equation (2.2) becomes

$$\mathbb{E}_t\langle \widehat{q}_{t+s}, f\rangle = \int_{(s,M]} f(u-s)\widehat{q}_t(du), \qquad \text{for } 0 < s < M.$$

At scheduled issuance dates, jump risk contributes no first-order risk premium. Specifically, the covariance term in Equation (2.7) is multiplied by a, so the first-order risk premium is determined by the conditional covariance between jumps in bond prices and portfolio value evaluated at the baseline price P_0. Since P_0 is independent of supply and does not jump at issuance, this covariance is zero. The first-order price coefficient $P_c^{(1)}$ therefore has zero conditional expected jump there, and the discounted pricing argument in Equation (2.13) applies unchanged between these dates. Taking the same small-volatility limit and removing the common factor $-a\sigma^2$ gives

$$\begin{aligned}
\widehat{X}_t(\tau) &= \int_0^\tau B_{\kappa_r}(\tau - s)\mathbb{E}_t\langle \widehat{q}_{t+s}, B_{\kappa_r}\rangle ds \\
&= \int_0^M \left[\int_0^{\min(\tau,u)} B_{\kappa_r}(\tau - s)B_{\kappa_r}(u-s)ds\right] \widehat{q}_t(du) \\
&= \langle \widehat{q}_t, H_{\kappa_r,0}(\tau,\cdot)\rangle.
\end{aligned} \tag{4.11}$$

Thus the price kernel is unchanged by fixing issuance times.

For a cohort with remaining maturity $u_i(t) = M - t + T_i$, independence of issue sizes gives,

for deterministic continuous f and g and $0 \le h < M$,

$$\mathrm{Cov}(\langle \widehat{q}_t, f\rangle, \langle \widehat{q}_{t+h}, g\rangle) = \mathrm{Var}(A_i) \sum_{i:\, h<u_i(t)\le M} f(u_i(t))g(u_i(t)-h).$$

This covariance depends on the position of t in the issuance cycle. As t runs through one interval of length Δ, the remaining maturities in the sum cover $(h, M]$ exactly once. Averaging over that interval therefore yields

$$\frac{1}{\Delta}\int_0^\Delta \mathrm{Cov}(\langle \widehat{q}_t, f\rangle, \langle \widehat{q}_{t+h}, g\rangle)dt = \frac{\mathrm{Var}(A_i)}{\Delta}\int_0^{M-h} f(u+h)g(u)du. \tag{4.12}$$

Equation (4.12) has the same form as Equation (3.2), with $\nu\mathbb{E}[A_i^2]$ replaced by $\mathrm{Var}(A_i)/\Delta$. In particular, the covariance operator of $\widehat{X}_t$ averaged over the issuance cycle is $[\mathrm{Var}(A_i)/\Delta]\mathcal{G}_{\kappa_r}^2$. Its principal component directions are therefore the same ϕ_n as in Section 3, and the corresponding scores are $\widehat{F}_{n,t} = \langle \phi_n, \widehat{X}_t\rangle = \eta_n\langle \widehat{q}_t, \phi_n\rangle$. By Equation (4.12), their average variances and autocorrelations based on the averaged covariances are

$$\begin{aligned} \overline{\mathrm{Var}}(\widehat{F}_n) &:= \frac{1}{\Delta}\int_0^\Delta \mathrm{Var}(\widehat{F}_{n,t})dt = \frac{\mathrm{Var}(A_i)}{\Delta}\eta_n^2, \\ \text{and}\quad \overline{\rho}_n(h) &:= \frac{\int_0^\Delta \mathrm{Cov}(\widehat{F}_{n,t}, \widehat{F}_{n,t+h})dt}{\int_0^\Delta \mathrm{Var}(\widehat{F}_{n,t})dt} = \int_0^{M-h} \phi_n(u+h)\phi_n(u)du. \end{aligned} \tag{4.13}$$

The kernel and the moment formulas in Equation (4.13) now coincide with those used in Theorems 1 and 2, apart from the variance multiplier. From this point onward, their proofs apply unchanged and give the following two propositions.

Proposition 6 (Persistence with deterministic issuance times). *Under the fixed issuance schedule above, for $\kappa_r \ge 0$, $\bar{r} = \lambda = 0$, and every n, the autocorrelation defined in Equation (4.13) satisfies*

$$\begin{aligned} \overline{\rho}_n'(0+) &= -\frac{\phi_n(M)^2}{2} \\ &= -\frac{2}{M} + \frac{\eta_n}{2M}\left[\kappa_r\phi_n'(M)^2 + 2\kappa_r^2\int_0^M \phi_n'(u)^2du\right] \in \left[-\frac{2}{M}, 0\right]. \end{aligned} \tag{4.14}$$

In particular, $\overline{\rho}_n'(0+) = -2/M$ for every component when $\kappa_r = 0$.

Proposition 7 (Variances of the principal components with deterministic issuance times). *Under the fixed issuance schedule above, suppose $\kappa_r = \bar{r} = \lambda = 0$. Let ζ_n be the positive roots of $\cos\zeta\cosh\zeta = -1$ in increasing order. Then $\eta_n = M^4/\zeta_n^4$ and $\zeta_n \sim (n - \frac{1}{2})\pi$,*

so the average variance defined in Equation (4.13) is

$$\overline{\mathrm{Var}}(\widehat{F}_n) = \frac{\mathrm{Var}(A_i)M^8}{\Delta\zeta_n^8} \sim \frac{\mathrm{Var}(A_i)M^8}{\Delta[(n-\frac{1}{2})\pi]^8}, \qquad \text{as } n \to \infty. \tag{4.15}$$

5 Numerical analysis

This section computes bond prices with stochastic short rates, demand, and issuance cohorts at finite risk aversion and volatility. I calibrate the model to moments from Treasury yields, trading-volume shares, and external estimates of demand sensitivity and marginal balance-sheet costs, and assess its fit to the data. Holding that cross-sectional fit fixed, I then isolate the cohort effect on the persistence of principal components.

5.1 Data

The yield data consist of 476 monthly Treasury zero-coupon curves from November 1985 through June 2025, constructed by Filipović, Pelger, and Ye (2026). To follow the calibration design of Vayanos and Vila (2021), I require thirty-year yields and therefore begin the sample when they first become available. Using this dataset, I calculate moments of yield levels and annual changes using monthly observations at annual maturities from one through thirty years.

Trading-volume data come from the Federal Reserve Bank of New York's *Primary Dealer Statistics* from April 2013 through June 2025. Following Vayanos and Vila (2021), I use nominal Treasury coupon transactions, excluding bills and inflation-protected securities. I compute the shares of total reported volume at remaining maturities of two years or less and above eleven years, which serve as target moments in the calibration.

The empirical issuance distribution uses 2,105 issues of nominal Treasury notes and bonds settling during 1985–2025, including reopenings.[6] For issue i, define

$$L_i = \frac{\text{days from issue to maturity}}{365.25}, \qquad \text{and} \quad A_i = \frac{Q_i}{\mathrm{GDP}_{t(i)}}, \tag{5.1}$$

where Q_i is accepted face amount and $\mathrm{GDP}_{t(i)}$ is nominal annual GDP in the issue year. The model uses the empirical joint distribution of (L_i, A_i), preserving the relation between issue size and maturity. A reopening uses its remaining term at settlement. Each issue is treated as a zero-coupon claim paying its normalized face amount at final maturity.

[6] The U.S. Treasury's *Treasury Securities Auctions Data* provides the auction records.

The maximum maturity and average number of issues per year are

$$M = \max_i L_i \approx 30.253 \text{ years}, \qquad \text{and} \quad \nu = 51.3 \text{ issues per year}.$$

Here M denotes only the maximum maturity observed in the issuance data; each issue has its own issuance maturity L_i. The issuance intensity ν is calculated by averaging annual issue counts over the sample period.

5.2 Nonlinear bond pricing

To determine bond prices, I first specify the functional forms of demand. To this end, I extend the demand profiles in Vayanos and Vila (2021) with a linear maturity term in the intercept and set the maturity cutoff to M:

$$\alpha(u) = \alpha e^{-\delta_\alpha u}\mathbf{1}_{\{u<M\}}, \qquad \text{and} \quad b(u,\beta) = (\theta_0 + \theta_1 u + \theta_2\beta)(e^{-\delta_\alpha u} - e^{-\delta_\theta u})\mathbf{1}_{\{u<M\}}. \tag{5.2}$$

Following Vayanos and Vila (2021), I impose $\delta_\theta > \delta_\alpha > 0$.

With heterogeneous issuance maturities, only issues with lifetime $L > u$ contribute to supply at remaining maturity u. The stationary mean and variance of supply, both per unit of remaining maturity, are

$$m_p(u) = \nu\mathbb{E}[A^p\mathbf{1}_{\{L>u\}}], \qquad \text{for } p = 1, 2.$$

For any maturity weighting function k, centered supply is

$$\langle q_t, k\rangle = \sum_{T_i \le t < T_i + L_i} A_i k(L_i - t + T_i) - \int_0^M m_1(u)k(u)du. \tag{5.3}$$

In addition, Equation (2.2) holds with heterogeneous issuance maturities.

Following the dollar-supply formulations of Greenwood and Vayanos (2014) and He, Nagel, and Song (2022), I specify the cohort component of net supply to arbitrageurs in dollar units. For the quantitative model, I assume that this component is valued at an equilibrium reference price, so market clearing is

$$P_{a,\lambda}(u)\Theta_a(du) = \bar{P}_{a,\lambda}(u)q(du) + D_a(u)du, \qquad \text{where} \quad \bar{P}_{a,\lambda}(u) = P_{a,\lambda}(\bar{r}, 0, 0, u). \tag{5.4}$$

The reference price $\bar{P}_{a,\lambda}(u)$ is the equilibrium bond price when the short rate equals $\bar{r}$ and both the demand factor β and centered supply q are zero. Thus arbitrageurs absorb $\bar{P}_{a,\lambda}(u)q(du)$ in dollar terms, while preferred-habitat investors absorb the remaining variation in face supply.

Therefore, arbitrageurs' face holdings associated with cohorts are $\bar{P}_{a,\lambda}(u)q(du)/P_{a,\lambda}(u)$. This is an assumption for isolating the effects of issuance and aging. It abstracts from the feedback of contemporaneous bond prices into the cohort component of dollar supply.

Under Equation (5.4), the pricing equation admits the affine representation

$$P_{a,\lambda}(r,\beta,q,\tau) = \exp\left[-A_{a,\lambda}(\tau) - \boldsymbol{B}_{a,\lambda}(\tau)^\top \boldsymbol{z} - \langle q, K_{a,\lambda}(\tau,\cdot)\rangle\right], \tag{5.5}$$

where $\boldsymbol{z} = (r-\bar{r},\beta)^\top$ and $\boldsymbol{B}_{a,\lambda} = (B_{a,\lambda,r}, B_{a,\lambda,\beta})^\top$. I solve for the coefficient functions jointly at finite a, λ, and σ, retaining the full empirical issuance-jump distribution. The coefficient functions satisfy

$$\begin{aligned}
A'_{a,\lambda}(\tau) &= \bar{r} - R(\tau) + \int_0^M [a\Omega(\tau,u)+\lambda] d_0(u)du, \\
\boldsymbol{B}'_{a,\lambda}(\tau) &= \boldsymbol{e}_1 - \boldsymbol{\Lambda}\boldsymbol{B}_{a,\lambda}(\tau) + \int_0^M [a\Omega(\tau,u)+\lambda]\boldsymbol{d}_a(u)du, \\
\text{and}\quad (\partial_\tau + \partial_u)K_{a,\lambda}(\tau,u) &= [a\Omega(\tau,u)+\lambda]e^{-A_{a,\lambda}(u)} \\
&\quad - \int_0^M [a\Omega(\tau,v)+\lambda]\alpha(v)K_{a,\lambda}(v,u)dv,
\end{aligned}$$

where $\boldsymbol{\Lambda} = \mathrm{diag}(\kappa_r,\kappa_\beta)$, $\boldsymbol{e}_i$ is the ith coordinate vector, $d_0(u) = b(u,0) - \alpha(u)A_{a,\lambda}(u)$, and $\boldsymbol{d}_a(u) = -\alpha(u)\boldsymbol{B}_{a,\lambda}(u) + \partial_\beta b(u,\beta)\boldsymbol{e}_2$. The functions $\Omega(\tau,u)$ and $R(\tau)$ denote the instantaneous covariance of bond returns and the convexity correction from expected log returns to expected returns, respectively. The boundary conditions are $A_{a,\lambda}(0) = 0$, $\boldsymbol{B}_{a,\lambda}(0) = \boldsymbol{0}$, and $K_{a,\lambda}(0,u) = K_{a,\lambda}(\tau,0) = 0$. Online Appendix E.1 defines Ω and R and provides a detailed derivation of these equations.

To assess the numerical accuracy of this price solution, I keep arbitrageurs' dollar holdings and their exposures to shocks fixed at the values implied by the solved prices at each state. I then recompute bond prices using a separate differential-equation calculation and a finer grid for the remaining maturities of issued bonds. Under the stationary distribution, the maximum root mean square (RMS) difference between the recomputed and original yields across maturities is 0.053 bp. Online Appendix E.2 provides further details.

The resulting bond prices imply the following covariances of yields. Let $\boldsymbol{C}$ and $\boldsymbol{\Gamma}(h)$ denote the covariance and lagged covariance matrices of yields at the observed maturities. For $0 \le h \le M$, the covariance between the yield at maturity τ_i at date t and the yield at maturity τ_j at date $t+h$ is

$$\Gamma_{ij}(h) = \sum_{\ell \in \{r,\beta\}} \frac{\sigma^2}{2\kappa_\ell \tau_i \tau_j} e^{-\kappa_\ell h} B_{a,\lambda,\ell}(\tau_i) B_{a,\lambda,\ell}(\tau_j) + \frac{1}{\tau_i \tau_j} \int_h^M m_2(u) K_{a,\lambda}(\tau_i, u) K_{a,\lambda}(\tau_j, u-h) du.$$

The first term comes from the short rate and demand factor, and the second from issuance cohorts outstanding at both dates. Online Appendix E.3 derives this expression.

5.3 Moment calculation for calibration

I follow the calibration design of Vayanos and Vila (2021). Specifically, I retain six yield moments: the one-year yield's level and annual-change standard deviations, the maturity averages of these two standard deviations, the average correlation of annual changes with the one-year yield, and the mean one-year yield. Maturity averages are taken over annual maturities from one through thirty years.

In addition, I add five moments. First, Vayanos and Vila (2021) use the trading-volume shares at maturities up to two years and above eleven years to identify δ_α and δ_θ, respectively. I include these shares as calibration moments. I compute the model shares by integrating $\mathbb{E}[((\partial_r D_a(u))^2 + (\partial_\beta D_a(u))^2)^{1/2}]$ over each range and dividing by its integral over $[0, M]$.[7] Next, using the demand response implied by the quantitative easing (QE) analysis of Vayanos and Vila (2021), I calibrate the sensitivity of preferred-habitat investors' demand so that a 1 bp increase in yields at all maturities raises dollar demand by 0.197% of GDP[8], imposing $10^{-4} \int_0^M u\alpha(u)du = 0.00197$. I also target the average level of the thirty-year yield and, following He, Nagel, and Song (2022) and Fleckenstein and Longstaff (2020), an average marginal balance-sheet cost of 81 bp/year, imposing $\lambda \mathbb{E}[\int_0^M P_{a,\lambda}(u)\Theta_a(du)] = 0.0081$.

I fix arbitrageurs' risk-aversion coefficient using $\gamma = aV$, where γ is relative risk aversion (RRA) and V is arbitrageur wealth measured in units of GDP (Vayanos and Vila, 2021). First, I set $\gamma = 5$ based on the evidence in Elminejad, Havranek, and Irsova (2025).[9] For V, I use

[7] Trading volume over a short interval is measured by the absolute change in preferred-habitat investors' dollar demand. With independent short-rate and demand shocks, its expected value is proportional to the square root of the sum of their variance contributions.

[8] Vayanos and Vila (2021) use the estimates of Krishnamurthy and Vissing-Jorgensen (2012) to obtain a demand increase of 0.59% of GDP for a 1 bp rise in yields at all maturities, corresponding to $\alpha = 5.21$. They treat this demand slope as an upper bound, with a corresponding lower bound $a = 6.78$. Their lower bound $\alpha = 1.04$ corresponds to the upper bound $a = 33.9$. For their QE analysis, they use $a = (6.78 + 33.9)/2 = 20.34$, which generates the average QE effect at the two bounds. Since their yield and trading-volume moments identify a and α only through their product $a\alpha$, tripling a from 6.78 to 20.34 requires reducing α to 5.21/3. For their fixed maturity profile, the demand increase for a 1 bp rise in yields is proportional to α, giving $0.59\%/3 \simeq 0.197\%$ of GDP.

[9] Their meta-analysis finds that, after correcting for publication bias, estimates of mean relative risk aversion in finance range from 2 to 7 across methods. They regard 5, the second most common calibration value after 10, as consistent with this evidence, whereas 10 lies above this range.

the SEC's *Private Fund Statistics, Second Calendar Quarter 2025*. The report gives net asset values of $309 billion for qualifying relative-value hedge funds, $207 billion for macro funds, and $237 billion for multi-strategy funds. These three categories have combined net assets of $753 billion.[10] Dividing by annualized nominal U.S. GDP of $30.486 trillion in 2025Q2 gives approximately 2.47%. Thus, setting $\gamma = 5$ gives $a = 5/0.0247 \simeq 202.428$, which I hold fixed in the calibration.

With a fixed, I jointly calibrate the remaining eleven parameters to these eleven moments by minimizing squared log moment differences:

$$\min_{\substack{\kappa_r,\sigma,\kappa_\beta,\alpha,\theta_2,\bar{r},\\ \theta_0,\delta_\alpha,\delta_\theta,\theta_1,\lambda}} \sum_{j=1}^{11} \left[\log \left(\frac{g_j^{\text{model}}}{g_j^{\text{target}}} \right) \right]^2, \tag{5.6}$$

where g_j denotes a target moment. After calibration, I refine the price solution at the calibrated parameters and recompute the moment discrepancies, as detailed in Online Appendix E.2.

5.4 Calibration results

Table 1 reports the model parameters and targeted moments. The calibrated $\kappa_r = 0.312$ and $\kappa_\beta = 0.071$ correspond to shock half-lives of about 2.2 and 9.8 years, respectively. Preferred-habitat demand conditions therefore adjust more slowly than the short rate. The average yields in the model are 3.451%, 3.929%, and 5.159% at one, seven, and thirty years, compared with 3.451%, 4.467%, and 5.159% in the data. The calibration captures the upward slope of the average yield curve.

Figure 6 shows patterns broadly similar to those in Vayanos and Vila (2021). Panels (a)–(c) are directly linked to the calibration targets, so their fit is partly achieved by construction. Panels (d)–(f) are not used in calibration. In panel (a), the standard deviation of yield levels generally declines with maturity in both the model and the data. The same pattern appears in panel (b) for the standard deviation of annual changes in yields. The model curves are relatively flat at longer maturities, with yield-level volatility remaining above the data at the long end. In panel (c), annual changes in yields at longer maturities are less correlated with annual changes in the one-year yield in both the model and the data. In panel (d), the model reproduces the broadly declining loadings of the first principal component of annual changes in yields, scaled to 1 for the 1-year maturity. This component explains 87.4% of the total variance of annual changes in yields across maturities in the model, compared with 85.2% in the data.

Panels (e)–(f) report the Fama and Bliss (1987) and Campbell and Shiller (1991) regression

[10] These strategies are prominent in Treasury repo markets (Banegas and Monin, 2023).

Table 1: Model parameters and targeted moments

Panel A: Model parameters		
Parameter	Description	Value
κ_r	Short-rate mean reversion	0.312
κ_β	Demand-factor mean reversion	0.071
σ	Short-rate diffusion scale	0.051
$\bar{r}$	Baseline short rate	0.026
a	Arbitrageur risk aversion (fixed)	202.428
θ_0	Mean demand adjustment	1.967
θ_1	Demand-intercept maturity slope	0.055
θ_2	Demand-factor loading	10.279
α	Demand slope	0.545
δ_α	Demand-slope decay rate	0.163
δ_θ	Demand-intercept decay rate	0.172
λ	Balance-sheet cost coefficient	0.943

Panel B: Targeted moments		
Moment	Target	Model
SD of one-year yield levels	2.553%p	2.555%p
SD of annual changes in the one-year yield	1.375%p	1.374%p
Average SD of yield levels	2.221%p	2.221%p
Average SD of annual yield changes	0.861%p	0.861%p
Average correlation with one-year yield changes	0.604	0.603
Mean of the one-year yield	3.451%	3.451%
Mean of the thirty-year yield	5.159%	5.159%
Trading-volume share: maturities up to two years	19.283%	19.329%
Trading-volume share: maturities above eleven years	11.347%	11.368%
Demand response to a 1 bp yield increase (% of GDP)	0.197%	0.197%
Mean marginal balance-sheet cost (bp/year)	81.000	81.000

Note: In the table, maturity averages are taken over annual maturities from one through thirty years. Holdings are measured relative to annual GDP.

coefficients. The former regresses the one-year excess bond return on the forward–spot spread; the latter regresses the annual yield change on the scaled yield spread. For maturity $n = 2, \ldots, 30$ years, the regressions take the following form:

$$p_{t+1}(n-1) - p_t(n) - y_t(1) = c_n^{\mathrm{FB}} + \gamma_n^{\mathrm{FB}}[f_t(n-1,n) - y_t(1)] + \epsilon_{t+1,n}^{\mathrm{FB}}, \tag{5.7}$$

$$\text{and} \quad y_{t+1}(n-1) - y_t(n) = c_n^{\mathrm{CS}} + \gamma_n^{\mathrm{CS}} \frac{y_t(n) - y_t(1)}{n-1} + \epsilon_{t+1,n}^{\mathrm{CS}}, \tag{5.8}$$

where $p_t(n)$ is the log price at time t of a bond with remaining maturity n years, and $f_t(n-1,n) = p_t(n-1) - p_t(n)$. Empirical coefficients are estimated from the yield data used for calibration. Model coefficients use the corresponding stationary covariances and variances.

The two regressions assess the connection between yield curve slopes and bond risk premia

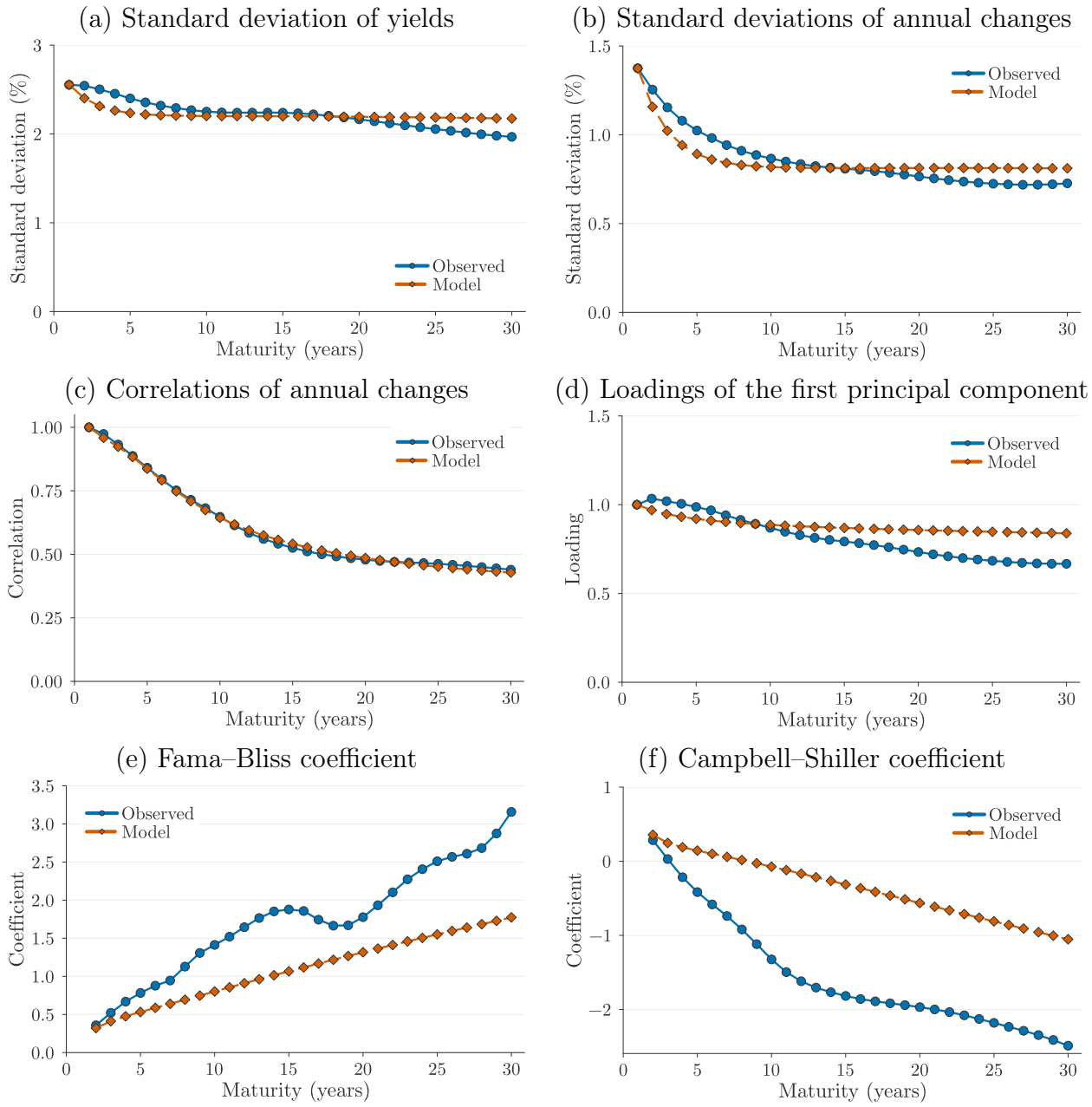


Figure 6: Comparison of theoretical and empirical moments

Notes: Panel (a) shows how the volatility of yield levels varies with maturity. Panel (b) shows how the volatility of annual yield changes varies with maturity. Panel (c) shows the correlation of annual yield changes at each maturity with the one-year yield change. Panel (d) shows the loadings of the first principal component of annual yield changes across maturities. Panel (e) reports the Fama and Bliss (1987) regression coefficients in Equation (5.7). Panel (f) reports the Campbell and Shiller (1991) regression coefficients in Equation (5.8). Theoretical moments use the calibrated model's stationary covariances and one-year lag covariances.

(Vayanos and Vila, 2021). Under the expectations hypothesis, $\gamma_n^{\mathrm{FB}} = 0$ and $\gamma_n^{\mathrm{CS}} = 1$. Positive Fama–Bliss coefficients and Campbell–Shiller coefficients below one instead indicate higher expected excess returns when the curve is steeper. Since these coefficients are not calibration targets, the comparison evaluates whether the fitted model also reproduces the predictable

variation in risk compensation observed in the data.

The regression results in panels (e)–(f) show that the model broadly reproduces the qualitative patterns in the data, but quantitative differences remain. Specifically, it generates positive Fama–Bliss coefficients that rise with maturity and negative Campbell–Shiller coefficients at long maturities. At thirty years, the model Fama–Bliss coefficient is 1.77, compared with 3.16 in the data, and the Campbell–Shiller coefficient is −1.05, compared with −2.49. Thus, both coefficients are smaller in magnitude than their empirical counterparts. Measured by deviations from the expectations-hypothesis values, predictability in the model is approximately 56% and 59% of its empirical counterpart in the Fama–Bliss and Campbell–Shiller regressions, respectively, at thirty years.[11]

5.5 Variance and autocorrelation of principal components in the calibrated model

Table 2 compares the standard deviations of the first nine principal components of yields in the data and model, together with their autocorrelations at lags of one, three, six, and twelve months. For the model, the table reports autocorrelations both with and without the cohort effect.

Table 2: Standard deviations and autocorrelations of principal components of yields in the data and model

	Data					Model								
							Autocorrelation (lag in months)							
	Standard deviation	Autocorrelation (lag in months)				Standard deviation	With cohort effect				Without cohort effect			
PC	(bp)	1	3	6	12	(bp)	1	3	6	12	1	3	6	12
1	1199.59	0.988	0.966	0.944	0.902	1195.44	0.994	0.982	0.965	0.931	0.993	0.981	0.964	0.930
2	213.59	0.982	0.938	0.864	0.657	226.58	0.976	0.929	0.860	0.732	0.877	0.832	0.770	0.659
3	43.01	0.882	0.694	0.522	0.200	30.67	0.988	0.943	0.842	0.606	0.001	0.001	0.001	0.001
4	31.07	0.934	0.823	0.697	0.557	16.20	0.977	0.896	0.721	0.360	0.001	0.001	0.001	0.001
5	18.88	0.894	0.782	0.654	0.399	10.04	0.976	0.891	0.708	0.344	0.001	0.001	0.001	0.001
6	11.13	0.852	0.652	0.484	0.278	6.60	0.975	0.884	0.691	0.318	0.000	0.000	0.000	0.000
7	7.76	0.833	0.638	0.447	0.172	4.84	0.979	0.901	0.730	0.382	0.000	0.000	0.000	0.000
8	5.24	0.808	0.599	0.443	0.055	3.59	0.981	0.898	0.708	0.305	0.000	0.000	0.000	0.000
9	3.05	0.679	0.432	0.185	−0.159	2.68	0.974	0.867	0.629	0.192	0.000	0.000	0.000	0.000

Note: PCA uses annual maturities from one through thirty years, separately for the data and model. Model moments are computed from the stationary covariances of yields.

[11] This combination of qualitative agreement and weaker return predictability is also reported by Vayanos and Vila (2021). At long maturities, predictability in their model is approximately 60% of its empirical counterpart.

The model PC1 and PC2 account for 96.441% and 3.464% of total yield variance, respectively. Together, these two principal components explain approximately 99.91% of the variance across the thirty annual maturities. These variance shares suggest approximating yields with PC1 and PC2 as factors and treating the higher principal components as pricing errors.

The economic interpretation of the two leading PCs also supports a two-factor approximation. To obtain this economic interpretation, I decompose their variances by regressing their standardized scores on standardized (r, β).[12] Since the short-rate, demand, and supply processes are independent, the squared coefficients measure the variance shares explained by the short-rate and demand factors, while the variance of the regression residual measures the contribution of the cohort effect. PC1 closely tracks the demand factor, which accounts for 99.5% of its variance. The short-rate factor and the cohort effect account for 89.5% and 10.0% of PC2's variance, respectively. For PC3–PC30, the cohort effect accounts for more than 99.8% of the variance of each component.

Thus, the short-rate and demand factors are mainly reflected in the two leading PCs. Because new issuance shocks are independent across dates, one might initially view these two PCs, which mainly reflect the short-rate and demand factors, as the only persistent sources of risk. In other words, PC1 and PC2 appear to exhaust systematic yield variation. One might then expect higher principal components to have small variances and low autocorrelations. Yet the model results with the cohort effect in Table 2 show that higher components, starting with PC3, exhibit strong autocorrelation despite their small variance shares. A low-dimensional common factor approximation can therefore explain almost all yield variance while leaving strongly autocorrelated residuals when the cohort effect is present.

If the strong autocorrelation of components beyond PC2 is due to the cohort effect, removing that effect should leave these components with low autocorrelation. To remove the cohort effect, I draw supply independently across dates from its stationary distribution. This breaks the link to the initial issues. However, I hold the calibrated parameters, price function, and short-rate and demand-factor dynamics fixed. Online Appendix E.3 describes in detail how the cohort effect is removed in the calculation. The model results without the cohort effect in Table 2 show that the autocorrelations of PC1 and PC2 remain largely unchanged or decline modestly. In contrast, those of all higher PCs fall to nearly zero.

The cohort effect therefore generates strong autocorrelation in higher principal components and in the residuals of a low-dimensional yield approximation. This persistence arises because issued bonds remain outstanding until maturity. Arbitrageurs require compensation for bearing the remaining interest-rate risk of these bonds, so the same issues continue to affect

[12] The regression coefficients are computed from the theoretical covariances between the PCs and (r, β) and the corresponding variances.

prices as they age toward redemption. When the cohort effect is removed, supply still affects prices, but no longer evolves from the initial supply state. This breaks the cohort link across dates while preserving the short-rate and demand-factor dynamics.

Table 2 also shows that higher principal components in the data generally have larger standard deviations and lower short-lag autocorrelations than in the model with the cohort effect. Specifically, the square root of the sum of variances of PC3–PC30 is 37.48 bp in the model and 58.38 bp in the data, so the former is 64.2% of the latter. The model therefore generates somewhat smaller residuals than those in the data. This difference in residual scale points to omitted factors or mechanisms beyond the cohort effect. The lower short-lag autocorrelations of higher PCs in the data suggest that these omitted components are less persistent than the cohort component.

5.6 Counterfactual analysis of relative risk aversion and balance-sheet costs

This section examines two counterfactuals to assess how relative risk aversion and balance-sheet costs influence the cohort effect. The first lowers RRA from $\gamma = 5$ to $\gamma = 2$, and the second removes balance-sheet costs by setting $\lambda = 0$. To isolate these effects, I hold all other parameters at their calibrated values and recompute equilibrium prices in each case.

Table 3 reports the results of lowering γ from 5 to 2. As RRA declines, a falls from

Table 3: Standard deviations and autocorrelations of principal components of yields at different levels of relative risk aversion

	$\gamma = 5$					$\gamma = 2$								
		Autocorrelation					Autocorrelation							
	Standard deviation	With cohort effect				Standard deviation	With cohort effect				Without cohort effect			
PC	(bp)	1	3	6	12	(bp)	1	3	6	12	1	3	6	12
1	1195.44	0.994	0.982	0.965	0.931	1168.64	0.994	0.982	0.964	0.930	0.992	0.981	0.963	0.929
2	226.58	0.976	0.929	0.860	0.732	192.60	0.976	0.929	0.860	0.733	0.880	0.835	0.773	0.662
3	30.67	0.988	0.943	0.842	0.606	25.56	0.987	0.932	0.806	0.525	0.000	0.000	0.000	0.000
4	16.20	0.977	0.896	0.721	0.360	13.95	0.975	0.880	0.681	0.298	0.001	0.001	0.001	0.001
5	10.04	0.976	0.891	0.708	0.344	8.63	0.975	0.884	0.693	0.330	0.000	0.000	0.000	0.000
6	6.60	0.975	0.884	0.691	0.318	5.71	0.974	0.876	0.673	0.290	0.000	0.000	0.000	0.000
7	4.84	0.979	0.901	0.730	0.382	4.17	0.981	0.904	0.736	0.396	0.000	0.000	0.000	0.000
8	3.59	0.981	0.898	0.708	0.305	3.09	0.981	0.889	0.682	0.271	0.000	0.000	0.000	0.000
9	2.68	0.974	0.867	0.629	0.192	2.35	0.968	0.842	0.573	0.100	0.000	0.000	0.000	0.000

Note: PCA uses annual maturities from one through thirty years, separately for each model specification. Model moments are computed from the stationary covariances of yields. Autocorrelation lags are in months. For $\gamma = 5$ and $\gamma = 2$, a equals 202.428 and 80.971, respectively. The eleven parameters other than a are held fixed at their baseline calibrated values, and equilibrium prices are recomputed.

202.428 to 80.971, with standard deviations decreasing for PC1–PC9 and autocorrelations generally declining as well. Since higher PCs primarily reflect the cohort effect, their smaller fluctuations and generally lower persistence indicate a weakening of this effect. Thus, lower RRA appears to weaken the cohort effect. The weakening of the cohort effect as RRA declines is also predicted by the theoretical approximation in Equation (4.7). The equation says that the magnitude of the cohort effect decreases as a falls. With lower risk aversion, arbitrageurs require less compensation to hold the same issuance cohort, so its effect on bond prices is smaller.

Next, I examine the role of balance-sheet costs. Section 4.2 shows that balance-sheet costs alter the variance and autocorrelation structure of the cohort effect. To examine these effects in the calibrated model, I set $\lambda = 0$ and resolve equilibrium prices with all other parameters fixed, then compare the resulting standard deviations and autocorrelations of the PCs in Table 4. The standard deviation of PC3 is nearly unchanged, but those of PC4–PC9 are smaller without balance-sheet costs. The residual scale falls from 37.48 to 35.03 bp, remaining 60.0% of its empirical counterpart. This indicates that balance-sheet costs slightly amplify the cohort effect. Regarding autocorrelation, Table 4 shows that each of the higher PCs (PC3–PC9) is more strongly autocorrelated at all four reported lags without balance-sheet costs than with them. Balance-sheet costs therefore dampen the persistence of these residual components. This stronger persistence without balance-sheet costs largely disappears when supply is redrawn independently across dates, with the one-month autocorrelations of PC3–

Table 4: Standard deviations and autocorrelations of principal components of yields with and without balance-sheet costs

	With balance-sheet costs					Without balance-sheet costs								
		Autocorrelation					Autocorrelation							
	Standard deviation	With cohort effect				Standard deviation	With cohort effect				Without cohort effect			
PC	(bp)	1	3	6	12	(bp)	1	3	6	12	1	3	6	12
1	1195.44	0.994	0.982	0.965	0.931	1147.38	0.994	0.981	0.963	0.927	0.993	0.981	0.962	0.927
2	226.58	0.976	0.929	0.860	0.732	531.55	0.975	0.926	0.858	0.737	0.970	0.922	0.854	0.733
3	30.67	0.988	0.943	0.842	0.606	31.81	0.989	0.966	0.927	0.840	0.003	0.003	0.003	0.002
4	16.20	0.977	0.896	0.721	0.360	12.14	0.988	0.960	0.908	0.776	0.001	0.001	0.001	0.001
5	10.04	0.976	0.891	0.708	0.344	6.24	0.989	0.959	0.896	0.731	0.000	0.000	0.000	0.000
6	6.60	0.975	0.884	0.691	0.318	3.99	0.991	0.963	0.899	0.718	0.000	0.000	0.000	0.000
7	4.84	0.979	0.901	0.730	0.382	2.53	0.990	0.956	0.875	0.646	0.000	0.000	0.000	0.000
8	3.59	0.981	0.898	0.708	0.305	1.79	0.991	0.955	0.865	0.606	0.000	0.000	0.000	0.000
9	2.68	0.974	0.867	0.629	0.192	1.22	0.989	0.946	0.838	0.543	0.000	0.000	0.000	0.000

Note: PCA uses annual maturities from one through thirty years, separately for each model specification. Model moments are computed from the stationary covariances of yields. Autocorrelation lags are in months. In the case without balance-sheet costs, equilibrium prices are recomputed with $\lambda = 0$ and all other parameters fixed at their calibrated values.

PC9 falling from the 0.988–0.991 range to nearly zero. Thus, residual persistence reflects the cohort effect both with and without balance-sheet costs.

Meanwhile, the standard deviation of PC2 rises from 226.58 to 531.55 bp without balance-sheet costs, as shown in Table 4. This is substantially higher than the 213.59 bp observed in the data. Since the share of the variance of PC2 explained by the short-rate factor rises from 89.5% to 97.3%, the rise in PC2 volatility appears to reflect a stronger influence of the short rate on PC2. To understand this stronger influence, consider Equation (2.7). The required return $\mathbb{E}_t[dP_{a,\lambda}(\tau)]/[P_{a,\lambda}(\tau)dt]$ includes the short rate and compensation for balance-sheet costs. At the calibrated parameters, a higher short rate reduces the aggregate market value of arbitrageurs' bond holdings by increasing preferred-habitat investors' aggregate demand for bonds. The resulting decline in balance-sheet compensation partially offsets the direct increase in required returns from the short rate. With $\lambda = 0$, this offset disappears, increasing the response of PC2 to short-rate shocks.

6 Discussion

The calibrated model's first two principal components capture almost all yield variance. The first closely tracks the demand factor, while the second reflects the short-rate factor and the cohort effect. Higher principal components, starting with PC3, have very small variances but strong autocorrelations due to the cohort effect. Residuals in the low-dimensional approximation reflect part of the continuing compensation investors require for bearing interest-rate risk from the same bond issues.

These results have two implications for specifying and evaluating bond pricing models. First, explaining almost all yield variance does not justify treating pricing residuals as serially independent measurement error. In Table 2, the model has the same contemporaneous fit with and without the cohort effect, but the autocorrelations of its residuals differ sharply. Improving the fit further need not resolve this problem. If an extended model captures an omitted factor independent of the cohort component while leaving the residual cohort component unchanged, its errors can become smaller but remain strongly autocorrelated, as more of the remaining variation comes from the cohort effect.

Second, residual autocorrelation does not by itself identify a missing macroeconomic factor. Even when an additional statistical factor reduces autocorrelation, it may be absorbing the cohort effect. The factor's persistence can then reflect the remaining lifetimes of bonds rather than an additional macroeconomic state. A macroeconomic interpretation therefore requires evidence that separates changes in the underlying economic state from the continued effects of outstanding bonds on prices.

The cohort effect also explains why bond issues of different maturities can have different

effects on prices even when they add the same amount of interest-rate risk to arbitrageurs' portfolios at issuance. For example, a large issue of two-year bonds can expose arbitrageurs to the same interest-rate risk today as a smaller issue of ten-year bonds. Two years later, the first issue has been redeemed, while the second still has eight years to maturity. Because prices today reflect the compensation arbitrageurs require for bearing risk until redemption, the two issues can have different price effects at issuance. The same logic applies to central bank purchases. A purchase reduces the supply that arbitrageurs must absorb for as long as the central bank holds the bonds, so its price effect depends on which maturities it buys and whether it holds the bonds to redemption or sells them earlier.

Finally, the model's contribution is to isolate the cohort effect, rather than to provide a complete account of empirical pricing residuals. Its residual yield variation is somewhat smaller than in the data, while its higher components generally have stronger short-lag autocorrelations. The quantitative contribution of the cohort effect to observed residuals therefore remains an empirical question. Measuring that contribution requires evidence on supply itself. Responses to identified issuance or purchase shocks, followed across remaining maturities as the affected bonds age, would test whether residual dynamics reflect the cohort effect and changing risk exposure.

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Online Appendix: Bonds Age, Pricing Errors Persist

A Auxiliary results and proofs

Demand and cohort components. Around $(a, \lambda) = (0, 0)$, the first-order price and log-price expansions are

$$\begin{aligned} P_{a,\lambda} &= P_0 + aP_d^{(1)} + aP_c^{(1)} + \lambda P_\lambda^{(1)} + o(a+\lambda), \\ \text{and} \quad \log P_{a,\lambda} &= \log P_0 + a\frac{P_d^{(1)}}{P_0} + a\frac{P_c^{(1)}}{P_0} + \lambda\frac{P_\lambda^{(1)}}{P_0} + o(a+\lambda). \end{aligned} \tag{A.1}$$

Here $P_d^{(1)}$ is the demand component, $P_c^{(1)}$ is the cohort component, and $P_\lambda^{(1)}$ is the balance-sheet cost coefficient. We now derive integral representations for $P_d^{(1)}$, $P_c^{(1)}$, and $P_\lambda^{(1)}$.

Define $D_0(r, \beta, u) = \alpha(u) \log P_0(r, u) + b(u, \beta)$ and $\mathcal{L}_\beta = -\kappa_\beta \beta \partial_\beta + \sigma^2 \partial_{\beta\beta}/2$. Since $\Theta_0 = q + (D_0/P_0)du$, the portfolio's short-rate exposure is

$$\int \Theta_0(du)\partial_r P_0(r, u) = \langle q, \partial_r P_0 \rangle + \int_0^M D_0(r, \beta, u)\partial_r \log P_0(r, u)du. \tag{A.2}$$

Since the covariance in Equation (2.7) is multiplied by a, its contribution at first order is determined by its value at $(a, \lambda) = (0, 0)$. The market-clearing condition $\theta_t = \Theta_0$ and the budget equation (2.6) give

$$\left.\frac{\mathrm{Cov}_t(dP_{a,\lambda}(\tau), dV_t)}{dt}\right|_{a=\lambda=0} = \sigma^2 \partial_r P_0(r, \tau) \int \Theta_0(du)\partial_r P_0(r, u). \tag{A.3}$$

Substituting Equation (A.2) into Equation (A.3) and matching the coefficients of a in Equation (2.7) gives

$$\begin{aligned} \partial_\tau (P_d^{(1)} + P_c^{(1)}) = &(\mathcal{L}_0 + \mathcal{L}_\beta - r)(P_d^{(1)} + P_c^{(1)}) \\ &- \sigma^2 \partial_r P_0(r, \tau)\langle q, \partial_r P_0 \rangle \\ &- \sigma^2 \partial_r P_0(r, \tau) \int_0^M D_0(r, \beta, u)\partial_r \log P_0(r, u)du, \end{aligned}$$

with both coefficients zero at redemption ($\tau = 0$). By linearity, the demand component satisfies

$$\begin{aligned} \partial_\tau P_d^{(1)} = &(\mathcal{L}_0 + \mathcal{L}_\beta - r)P_d^{(1)} \\ &- \sigma^2 \partial_r P_0(r, \tau) \int_0^M D_0(r, \beta, u)\partial_r \log P_0(r, u)du, \end{aligned}$$

with $P_d^{(1)}(r, \beta, \tau) = 0$ at redemption ($\tau = 0$). Solving this equation gives

$$P_d^{(1)}(r,\beta,\tau) = -\sigma^2 \mathbb{E}_{r,\beta} \int_0^\tau e^{-\int_0^s r_\ell d\ell} \partial_r P_0(r_s, \tau - s) \times \left[\int_0^M D_0(r_s, \beta_s, u) \partial_r \log P_0(r_s, u) du \right] ds.$$

With $\alpha(u)$ bounded and $b(u,\beta)$ affine in β with coefficients bounded in u, Gaussian finite-horizon moments make this integral admissible. It has zero payoff at redemption and does not depend on q, so $\mathcal{L}_q P_d^{(1)} = 0$.

The cohort coefficient solves Equation (2.10). Integrating its discounted equation to redemption gives Equation (2.13):

$$P_c^{(1)}(r,q,\tau) = -\sigma^2 \int_0^\tau \mathbb{E}_{r,q} \left[e^{-\int_0^s r_\ell d\ell} \partial_r P_0(r_s, \tau - s) \langle q_s, \partial_r P_0(r_s, \cdot) \rangle \right] ds.$$

Finally, we derive the balance-sheet cost coefficient $P_\lambda^{(1)}$. Substituting the market-clearing holdings $\Theta_0 = q + (D_0/P_0)du$ into Equation (2.7) and matching the coefficient of λ gives

$$\partial_\tau P_\lambda^{(1)} = (\mathcal{L}_0 + \mathcal{L}_\beta - r) P_\lambda^{(1)} - P_0(r,\tau) \left[\langle q, P_0(r,\cdot) \rangle + \int_0^M D_0(r,\beta,u) du \right],$$

with zero value at redemption. Discounting and integrating to redemption gives $P_\lambda^{(1)} = P_{\lambda,d}^{(1)} + P_{\lambda,c}^{(1)}$, where

$$P_{\lambda,d}^{(1)}(r,\beta,\tau) = -\int_0^\tau \mathbb{E}_{r,\beta} \left[e^{-\int_0^s r_\ell d\ell} P_0(r_s, \tau - s) \int_0^M D_0(r_s, \beta_s, u) du \right] ds,$$

$$\text{and} \quad P_{\lambda,c}^{(1)}(r,q,\tau) = -\int_0^\tau \mathbb{E}_{r,q} \left[e^{-\int_0^s r_\ell d\ell} P_0(r_s, \tau - s) \langle q_s, P_0(r_s, \cdot) \rangle \right] ds.$$

Using independence of the short rate and issuance and Equation (2.2), we rewrite the cohort component as

$$P_{\lambda,c}^{(1)}(r,q,\tau) = -\int_0^M \left[\int_0^{\min(\tau,u)} \mathbb{E}_r \left[e^{-\int_0^s r_\ell d\ell} P_0(r_s, \tau - s) P_0(r_s, u - s) \right] ds \right] q(du). \tag{A.4}$$

Thus $P_{\lambda,d}^{(1)}$ depends only on (r,β), while $P_{\lambda,c}^{(1)}$ is linear in centered supply. Since q_t is independent of (r_t, β_t) and has mean zero, $\mathbb{E}[P_{\lambda,c}^{(1)}/P_0 \mid r_t, \beta_t] = 0$. The same conditional-mean property holds for $P_c^{(1)}/P_0$, so the demand and cohort components have zero cross-covariance, including both balance-sheet cost contributions.

We conclude this section with two auxiliary lemmas. The first shows that the coefficient $P_c^{(1)}$ used in Proposition 1 is well defined. The second provides the results for the operator

needed for the principal component decomposition and the proof of Theorem 1.

Lemma A.1 (Admissibility and portfolio optimality). *For $\kappa_r \geq 0$, $\bar{r} \geq 0$, and fixed $\sigma > 0$, Equation (2.10) has a unique admissible solution given by Equation (2.13). The solution is linear in current net supply. At each pricing state (r, q), the market-clearing position $\theta = q$ maximizes the coefficient of a in the arbitrageur's objective over fixed finite face value holdings.*

Proof. Let $F_\sigma(s)$ denote the discounted integrand inside the expectation in Equation (2.13). To establish admissibility, we first bound its conditional second moment $\mathbb{E}_{r,q}|F_\sigma(s)|^2$. For fixed $\bar{r} \geq 0$ and $0 < \sigma \leq \bar{\sigma} < \infty$, Equation (2.9) bounds P_0 and its short-rate derivatives by $Ce^{M|r_s|}$. In addition, the cumulative face amount of new issuance follows a compound Poisson process with $\mathbb{E}[A^2] < \infty$. Therefore, the supply stock satisfies $\sup_{s \leq \tau} \mathbb{E}_q \|q_s\|_{\mathrm{TV}}^2 < \infty$. Independence of the short rate and issuance then yields

$$\mathbb{E}_{r,q}|F_\sigma(s)|^2 \leq C\mathbb{E}_q\|q_s\|_{\mathrm{TV}}^2 \, \mathbb{E}_r\left[e^{-2\int_0^s r_\ell d\ell + 4M|r_s|}\right].$$

Since $(r_s, \int_0^s r_\ell d\ell)$ is jointly Gaussian, the last expectation is bounded uniformly in s and σ. Hence

$$\sup_{\substack{0\leq s\leq \tau \\ 0<\sigma\leq\bar{\sigma}}} \mathbb{E}_{r,q}|F_\sigma(s)|^2 < \infty, \qquad \text{and} \qquad \int_0^\tau \mathbb{E}_{r,q}|F_\sigma(s)|ds < \infty. \tag{A.5}$$

Thus Equation (2.13) is finite, and uniform integrability justifies the limit in Proposition 1.

To verify the pricing equation, split Equation (2.13) at $0 < \epsilon < \tau$. The Markov property gives

$$P_c^{(1)}(r, q, \tau) = \mathbb{E}_{r,q}\left[e^{-\int_0^\epsilon r_s ds} P_c^{(1)}(r_\epsilon, q_\epsilon, \tau - \epsilon)\right] - \sigma^2 \int_0^\epsilon \mathbb{E}_{r,q} F_\sigma(s) ds.$$

Rearranging, dividing by ϵ, and letting $\epsilon \downarrow 0$ gives Equation (2.10); the empty integral at $\tau = 0$ gives its terminal condition.

To check admissibility, apply Itô's formula to the expression inside the expectation in the preceding equation and use Equation (2.10). Integrating from 0 to s gives

$$\begin{aligned} e^{-\int_0^s r_\ell d\ell} P_c^{(1)}(r_s, q_s, \tau - s) &= P_c^{(1)}(r, q, \tau) + \sigma^2 \int_0^s F_\sigma(v) dv \\ &+ \sigma \int_0^s e^{-\int_0^v r_\ell d\ell} \partial_r P_c^{(1)}(r_v, q_{v-}, \tau - v) dW_v^r \\ &+ \int_0^s \int_0^\infty e^{-\int_0^v r_\ell d\ell} \left[P_c^{(1)}(r_v, q_{v-} + A\delta_M, \tau - v) - P_c^{(1)}(r_v, q_{v-}, \tau - v)\right] \widetilde{N}(dv, dA). \end{aligned}$$

Here q_{v-} is supply just before time v, and $\widetilde{N}(dv, dA)$ is the compensated Poisson random measure associated with issuance. The argument used above to bound $\mathbb{E}_{r,q}|F_\sigma(s)|^2$ also bounds

the conditional second moments of the integrands of the W^r and $\widetilde{N}$ integrals; the issuance jump is linear in A. Both stochastic integrals are therefore square integrable martingales with zero conditional mean. Together with the integrability of the drift integral, this verifies admissibility.

For uniqueness, the difference between two admissible solutions has zero discounted drift and a true martingale part. Their common terminal condition therefore implies

$$\begin{aligned}&(P_c^{(1)} - \widetilde{P}_c^{(1)})(r, q, \tau)\\ &\quad = \mathbb{E}_{r,q}\left[e^{-\int_0^\tau r_s ds}(P_c^{(1)} - \widetilde{P}_c^{(1)})(r_\tau, q_\tau, 0)\right] = 0.\end{aligned}$$

For linearity, independence and Equation (2.2) give

$$\mathbb{E}[\langle q_s, \partial_r P_0(r_s, \cdot)\rangle \mid (r_v)_{0\le v\le\tau}, q_0 = q] = \int_{(s,M]} \partial_r P_0(r_s, u-s) q(du).$$

Substituting into Equation (2.13) shows that the solution is linear in current q.

To verify portfolio optimality, hold (r, q) fixed and vary θ. At baseline prices the portfolio diffusion loading is $\sigma\langle\theta, \partial_r P_0\rangle$. The variance term multiplying a in the arbitrageur's objective is therefore $-\sigma^2\langle\theta, \partial_r P_0\rangle^2/2$. Equations (A.1) (with $\lambda = 0$ and $P_d^{(1)} = 0$) and (2.10) give $\sigma^2\partial_r P_0(r, u)\langle q, \partial_r P_0\rangle$ as the coefficient of a in the excess drift of a bond with remaining maturity u. Integrating this expression against $\theta(du)$ gives the drift term multiplying a in the arbitrageur's objective. After dividing by dt, the coefficient of a in the arbitrageur's objective is

$$\mathcal{J}_1(\theta; q) = \sigma^2\langle\theta, \partial_r P_0\rangle\langle q, \partial_r P_0\rangle - \frac{\sigma^2}{2}\langle\theta, \partial_r P_0\rangle^2.$$

Completing the square yields

$$\mathcal{J}_1(\theta; q) - \mathcal{J}_1(q; q) = -\frac{\sigma^2}{2}\langle\theta - q, \partial_r P_0\rangle^2 \le 0. \qquad \square$$

Lemma A.2 (Differential representation of the integral operator). *For $\kappa_r \ge 0$ and $\bar{r} = 0$, the operator $\mathcal{G}_{\kappa_r}$ is compact, self-adjoint, and strictly positive on $L^2(0, M)$. Its eigenfunctions are smooth and satisfy Equation (3.8) and the boundary conditions.*

Proof. Define the Volterra operator and its adjoint by

$$(\mathcal{A}_{\kappa_r} f)(\tau) = \int_0^\tau B_{\kappa_r}(\tau - s) f(s) ds, \qquad \text{and} \quad (\mathcal{A}^*_{\kappa_r} f)(s) = \int_s^M B_{\kappa_r}(u - s) f(u) du.$$

Changing the integration order gives $\mathcal{G}_{\kappa_r} = \mathcal{A}_{\kappa_r}\mathcal{A}^*_{\kappa_r}$. Its continuous symmetric kernel $H_{\kappa_r,0}$ implies compactness and self-adjointness.

If $w = \mathcal{A}^*_{\kappa_r} f$, differentiation gives

$$w'' - \kappa_r w' = f, \qquad \text{and} \quad w(M) = w'(M) = 0.$$

Thus $\mathcal{A}^*_{\kappa_r} f = 0$ implies $f = 0$, and

$$\langle f, \mathcal{G}_{\kappa_r} f \rangle = \|\mathcal{A}^*_{\kappa_r} f\|^2 > 0 \qquad (f \neq 0).$$

For $g = \mathcal{G}_{\kappa_r} f = \mathcal{A}_{\kappa_r} w$, differentiation also gives

$$g'' + \kappa_r g' = w, \qquad \text{and} \quad g(0) = g'(0) = 0.$$

Applying $d^2/d\tau^2 - \kappa_r\, d/d\tau$ to both sides of the preceding differential equation yields $g'''' - \kappa_r^2 g'' = f$. At M, the terminal conditions for w imply $g''(M) + \kappa_r g'(M) = 0$ and $g'''(M) + \kappa_r g''(M) = 0$, hence $g'''(M) - \kappa_r^2 g'(M) = 0$. Set $g = \eta_n \phi_n$ and $f = \phi_n$ to obtain Equation (3.8). The differential equation then gives smoothness. □

B Proof of Proposition 2

Proposition 2 (Persistence with discounting). *Let $F_{n,t} = \langle \phi_n, X_t \rangle$ denote the principal component scores, and let ρ_n denote their autocorrelations. For $\bar{r} \geq 0$, $\kappa_r \geq 0$, and every fixed n,*

$$-\frac{4}{M} \leq \rho_n'(0+) = d_n \leq 0, \qquad \textit{where} \quad d_n := -\frac{\psi_n(M)^2}{2}, \tag{4.4}$$

and

$$\rho_n(h) = 1 + d_n h + O((h/M)^2), \qquad \textit{as } h \downarrow 0. \tag{4.5}$$

Proof. Define

$$(J_\xi f)(u) = \int_0^u e^{-\xi(u-\ell)} f(\ell) d\ell,$$

$$\text{and} \quad (J^*_\xi f)(u) = \int_u^M e^{-\xi(\ell-u)} f(\ell) d\ell, \qquad \text{for } \xi \geq 0.$$

These operators solve, respectively,

$$z' + \xi z = f, \quad \text{with } z(0) = 0, \qquad \text{and} \quad -z' + \xi z = f, \quad \text{with } z(M) = 0. \tag{B.1}$$

Convolving their kernels gives

$$\mathcal{G} = J_0 J_{\kappa_r} J^*_{\kappa_r + \bar{r}} J^*_{\bar{r}}. \tag{B.2}$$

The continuous pricing kernel makes $\mathcal{G}$ compact. Each operator and its adjoint are injective,

so both $\mathcal{G}$ and $\mathcal{G}^*$ are injective. Thus $\mathcal{G}\mathcal{G}^*$ has a complete orthonormal family of eigenfunctions ϕ_n, with eigenvalues $s_n^2 > 0$ in nonincreasing order. Defining $\psi_n = \mathcal{G}^*\phi_n/s_n$ establishes Equation (4.2) with complete orthonormal singular families. Repeated application of the singular equations gives smooth functions with $\phi_n(0) = \psi_n(0) = 0$.

Equation (4.1) gives the covariance operator of X_t as $\nu\mathbb{E}[A_i^2]\mathcal{G}\mathcal{G}^*$. Equation (4.3) gives the PC scores. Setting $f = g = \phi_n$ in Equation (4.1) and using Equation (4.2) gives

$$\mathrm{Var}(F_{n,t}) = \nu\mathbb{E}[A_i^2]s_n^2. \tag{B.3}$$

Applying Equation (3.2) and differentiating from the right gives

$$\rho_n(h) = \int_0^{M-h} \psi_n(u+h)\psi_n(u)du, \qquad \text{for } 0 \le h < M, \tag{B.4}$$

$$\text{and} \quad \rho_n'(0+) = -\psi_n(M)^2 + \int_0^M \psi_n'\psi_n\, du = -\frac{\psi_n(M)^2}{2}. \tag{B.5}$$

For each fixed n, Taylor expansion of the smooth function ρ_n in Equation (B.4) gives

$$\rho_n(h) = 1 - \frac{\psi_n(M)^2}{2}h + O(h^2), \qquad \text{as } h \downarrow 0.$$

To obtain a lower bound on $\rho_n'(0+) = -\psi_n(M)^2/2$, let $\mathcal{E}_\xi(f) = \langle f, J_\xi f\rangle$. With $z = J_\xi f$, Equation (B.1) and integration by parts give

$$\mathcal{E}_\xi(f) = \frac{z(M)^2}{2} + \xi\int_0^M z(u)^2 du \ge 0. \tag{B.6}$$

Differentiating the kernel of J_ξ gives $\partial_\xi J_\xi = -J_\xi^2$. The operators J_ξ commute with each other, as do their adjoints. Equation (B.2) therefore yields

$$\partial_{\kappa_r}\mathcal{G} = -J_{\kappa_r}\mathcal{G} - \mathcal{G}J_{\kappa_r+\bar{r}}^*, \qquad \text{and} \quad \partial_{\bar{r}}\mathcal{G} = -\mathcal{G}(J_{\bar{r}}^* + J_{\kappa_r+\bar{r}}^*).$$

Pairing the preceding equations with the fixed singular functions in Equation (4.2) gives

$$\begin{aligned} \langle\phi_n, (\partial_{\kappa_r}\mathcal{G})\psi_n\rangle &= -s_n\{\mathcal{E}_{\kappa_r}(\phi_n) + \mathcal{E}_{\kappa_r+\bar{r}}(\psi_n)\}, \\ \text{and} \quad \langle\phi_n, (\partial_{\bar{r}}\mathcal{G})\psi_n\rangle &= -s_n\{\mathcal{E}_{\bar{r}}(\psi_n) + \mathcal{E}_{\kappa_r+\bar{r}}(\psi_n)\}. \end{aligned} \tag{B.7}$$

Differentiating the scaling identity $H_{\kappa_r,\bar{r}}(c\tau, cu) = c^3 H_{c\kappa_r,c\bar{r}}(\tau, u)$ with respect to c and then setting $c = 1$ gives

$$(\tau\partial_\tau + u\partial_u)H_{\kappa_r,\bar{r}} = 3H_{\kappa_r,\bar{r}} + \kappa_r\partial_{\kappa_r}H_{\kappa_r,\bar{r}} + \bar{r}\partial_{\bar{r}}H_{\kappa_r,\bar{r}}.$$

Multiplying both sides of the preceding equation by $\phi_n(\tau)\psi_n(u)$ and integrating over $(0, M)^2$ gives

$$\begin{aligned}\int_0^M \int_0^M \phi_n(\tau)\psi_n(u)(\tau\partial_\tau + u\partial_u)H_{\kappa_r,\bar{r}}(\tau, u)\, du\, d\tau \\ = 3\langle\phi_n, \mathcal{G}\psi_n\rangle + \kappa_r\langle\phi_n, (\partial_{\kappa_r}\mathcal{G})\psi_n\rangle \\ + \bar{r}\langle\phi_n, (\partial_{\bar{r}}\mathcal{G})\psi_n\rangle.\end{aligned} \tag{B.8}$$

Substituting the τ- and u-derivatives of Equation (4.2) into the left side of Equation (B.8) and integrating by parts gives

$$s_n \int_0^M u(\phi_n\phi_n' + \psi_n\psi_n')du = \frac{s_n}{2}\{M[\phi_n(M)^2 + \psi_n(M)^2] - 2\}. \tag{B.9}$$

Using $\langle\phi_n, \mathcal{G}\psi_n\rangle = s_n$ and Equation (B.7), the right side of Equation (B.8) becomes

$$s_n\{3 - \kappa_r\mathcal{E}_{\kappa_r}(\phi_n) - \bar{r}\mathcal{E}_{\bar{r}}(\psi_n) - (\kappa_r + \bar{r})\mathcal{E}_{\kappa_r+\bar{r}}(\psi_n)\}. \tag{B.10}$$

Equating the expressions in Equations (B.9) and (B.10) and rearranging gives

$$\begin{aligned}\frac{M}{2}[\phi_n(M)^2 + \psi_n(M)^2] = 4 - \kappa_r\mathcal{E}_{\kappa_r}(\phi_n) - \bar{r}\mathcal{E}_{\bar{r}}(\psi_n) \\ - (\kappa_r + \bar{r})\mathcal{E}_{\kappa_r+\bar{r}}(\psi_n).\end{aligned} \tag{B.11}$$

All subtracted terms are nonnegative by Equation (B.6). Hence $\psi_n(M)^2/2 \le 4/M$, which, together with Equation (B.5), proves Equation (4.4). □

C Proof of Proposition 3

Proposition 3 (Variances of the principal components with discounting). *Suppose $\kappa_r = 0$ and $\bar{r} \ge 0$. Define $\zeta_n > 0$ by $\zeta_n^2[\zeta_n^2 + (\bar{r}M)^2] = M^4/s_n$, ordered increasingly and counting multiplicity. Then,*

$$\mathrm{Var}(F_{n,t}) = \frac{\nu\mathbb{E}[A_i^2]M^8}{\zeta_n^4[\zeta_n^2 + (\bar{r}M)^2]^2} \sim \frac{\nu\mathbb{E}[A_i^2]M^8}{(\pi n)^8}, \qquad \text{and} \quad \zeta_n \sim \pi n. \tag{4.6}$$

The asymptotics hold as $n \to \infty$ for fixed M and $\bar{r}$.

Proof. Substituting $\kappa_r = 0$ into Equation (B.2) gives $\mathcal{G} = J_0^2(J_{\bar{r}}^*)^2$. By Equation (B.1), $z = J_\xi f$ implies $f = (\partial_u + \xi)z$ with $z(0) = 0$, while $z = J_\xi^* f$ implies $f = (-\partial_u + \xi)z$ with

$z(M) = 0$. Inverting reverses the order of composition, so

$$\mathcal{Q} = \mathcal{G}^{-1} = (-\partial_u + \bar{r})^2 \partial_u^2 = (\partial_u - \bar{r})^2 \partial_u^2,$$
$$\text{and} \quad \mathcal{Q}^* = (\mathcal{G}^*)^{-1} = (-\partial_u)^2 (\partial_u + \bar{r})^2 = \partial_u^2 (\partial_u + \bar{r})^2.$$

The domains consist of functions with square-integrable derivatives through order four, with endpoint conditions determined by the integral factors. Applying these inverses to Equation (4.2) gives

$$(\partial_u - \bar{r})^2 \partial_u^2 \phi_n = s_n^{-1} \psi_n, \qquad \text{and} \quad \partial_u^2 (\partial_u + \bar{r})^2 \psi_n = s_n^{-1} \phi_n. \tag{C.1}$$

The boundary conditions for ϕ_n and ψ_n are

$$\begin{gathered} \phi_n(0) = \phi_n'(0) = \psi_n(0) = \psi_n'(0) = 0, \qquad \phi_n''(M) = \phi_n'''(M) = 0, \\ \psi_n''(M) + 2\bar{r}\psi_n'(M) + \bar{r}^2 \psi_n(M) = 0, \\ \text{and} \quad \psi_n'''(M) + 2\bar{r}\psi_n''(M) + \bar{r}^2 \psi_n'(M) = 0. \end{gathered} \tag{C.2}$$

First, in the first relation of Equation (B.1), set $z = J_\xi^2 g$ and $f = J_\xi g$. Since $f(0) = 0$, this gives $z(0) = z'(0) = 0$. Applying this result with $\xi = 0$ and $\xi = \bar{r}$ to Equation (4.2), using the factorization in Equation (B.2), gives the conditions at zero.

Second, applying ∂_u^2 to the first relation in Equation (4.2) and using Equations (B.1) and (B.2) gives $s_n \phi_n'' = (J_{\bar{r}}^*)^2 \psi_n$. In the second relation of Equation (B.1), set $\xi = \bar{r}$, $z = (J_{\bar{r}}^*)^2 \psi_n$, and $f = J_{\bar{r}}^* \psi_n$. Since $f(M) = 0$, this yields $z(M) = z'(M) = 0$; combining these values with $z = s_n \phi_n''$ gives the conditions for ϕ_n at M.

Third, applying $(\partial_u + \bar{r})^2$ to the second relation in Equation (4.2) and using Equations (B.1) and (B.2) gives $s_n (\partial_u + \bar{r})^2 \psi_n = (J_0^*)^2 \phi_n$. In the second relation of Equation (B.1), set $\xi = 0$, $z = (J_0^*)^2 \phi_n$, and $f = J_0^* \phi_n$. Since $f(M) = 0$, this yields $z(M) = z'(M) = 0$; combining these values with $z = s_n (\partial_u + \bar{r})^2 \psi_n$ gives the conditions for ψ_n at M.

We have thus derived Equations (C.1) and (C.2) from Equation (4.2). Conversely, solving Equation (C.1) under the boundary conditions in Equation (C.2) by repeated application of Equation (B.1) recovers Equation (4.2). Hence both formulations give the same values of s_n.

Combining the two relations in Equation (C.1) to eliminate their common ϕ_n gives

$$\mathcal{Q}\mathcal{Q}^* \psi_n = \partial_u^4 (\partial_u^2 - \bar{r}^2)^2 \psi_n = s_n^{-2} \psi_n. \tag{C.3}$$

Here ψ_n satisfies its four conditions in Equation (C.2), and $\mathcal{Q}^* \psi_n$ satisfies the four conditions for ϕ_n by Equation (C.1). Equation (C.3) shows that the eigenvalues of $\mathcal{P} = \mathcal{Q}\mathcal{Q}^*$ are precisely s_n^{-2}.

Evaluating the differential polynomial in Equation (C.3) at $\partial_u = i\zeta_n/M$ gives

$$s_n^{-2} = \frac{\zeta_n^4[\zeta_n^2 + (\bar{r}M)^2]^2}{M^8}.$$

The eigenvalues s_n^{-2} are determined by Equations (C.1) and (C.2). For each such eigenvalue, the preceding equation uniquely determines $\zeta_n > 0$, because its right-hand side is strictly increasing from zero to infinity. Substituting the preceding equation into Equation (B.3) gives the exact variance formula in Equation (4.6).

To obtain the asymptotic behavior of s_n^{-2}, adapt the boundary-condition comparison argument in Yafaev (2015) to the eighth-order operator $\mathcal{P}$. Let $\mathcal{P}_{\text{per}}$ have the differential expression in Equation (C.3) with periodic boundary conditions $f^{(k)}(0) = f^{(k)}(M)$ for $k = 0, \ldots, 7$. The functions $e^{2\pi i j u/M}$ satisfy these conditions and can be chosen as eigenfunctions of $\mathcal{P}_{\text{per}}$. Since

$$\partial_u^k e^{2\pi i j u/M} = \left(\frac{2\pi i j}{M}\right)^k e^{2\pi i j u/M}, \qquad \text{for } k = 0, 1, \ldots,$$

Equation (C.3) gives their eigenvalues as

$$\left(\frac{2\pi j}{M}\right)^4 \left[\left(\frac{2\pi j}{M}\right)^2 + \bar{r}^2\right]^2, \qquad \text{for } j \in \mathbb{Z}.$$

The frequencies j and $-j$ have the same eigenvalue, so their rank satisfies $n \sim 2|j|$. Thus the increasingly ordered eigenvalues are asymptotic to $(\pi n/M)^8$.

To isolate the effect of the endpoint conditions, compare solutions for the same input $f \in L^2(0, M)$:

$$(\mathcal{P} + \mathrm{Id})z = f, \qquad \text{and} \quad (\mathcal{P}_{\text{per}} + \mathrm{Id})z_{\text{per}} = f,$$

where Id denotes the identity operator and each solution satisfies its respective endpoint conditions. The operator $\mathcal{P}$ is positive and self-adjoint, with compact inverse $\mathcal{G}^*\mathcal{G}$. Adding the identity Id shifts the zero periodic eigenvalue to one and makes $\mathcal{P} + \mathrm{Id}$ and $\mathcal{P}_{\text{per}} + \mathrm{Id}$ invertible. Hence,

$$z - z_{\text{per}} = [(\mathcal{P} + \mathrm{Id})^{-1} - (\mathcal{P}_{\text{per}} + \mathrm{Id})^{-1}]f.$$

Because the differential expressions agree, this difference satisfies $[\partial_u^4(\partial_u^2 - \bar{r}^2)^2 + \mathrm{Id}](z - z_{\text{per}}) = 0$. The homogeneous eighth-order equation has an eight-dimensional solution space, so $(\mathcal{P} + \mathrm{Id})^{-1} - (\mathcal{P}_{\text{per}} + \mathrm{Id})^{-1}$ has rank at most eight.

Under the shift and inversion, any eigenvalue μ of $\mathcal{P}$ or $\mathcal{P}_{\text{per}}$ becomes $(\mu + 1)^{-1}$. Applying the min–max principle to the compact self-adjoint inverses $(\mathcal{P} + \mathrm{Id})^{-1}$ and $(\mathcal{P}_{\text{per}} + \mathrm{Id})^{-1}$ and translating back to the original eigenvalues gives

$$\mu^{\text{per}}_{n-8} \leq s_n^{-2} \leq \mu^{\text{per}}_{n+8}, \qquad \text{for } n > 8,$$

where μ^{per}_n are the increasingly ordered eigenvalues of $\mathcal{P}_{\text{per}}$, counting multiplicity. This finite shift preserves the leading asymptotic, giving $s_n^{-2} \sim (\pi n/M)^8$ and hence $s_n \to 0$. By the defining relation $\zeta_n^2[\zeta_n^2 + (\bar{r}M)^2] = M^4/s_n$ in Proposition 3, $s_n \to 0$ implies $\zeta_n \to \infty$ for fixed M and $\bar{r}$. Squaring the same relation therefore gives

$$s_n^{-2} = \frac{\zeta_n^8}{M^8}\left[1 + \frac{(\bar{r}M)^2}{\zeta_n^2}\right]^2 \sim \frac{\zeta_n^8}{M^8}.$$

Here the second term inside the square brackets, $(\bar{r}M)^2/\zeta_n^2$, tends to zero as $\zeta_n \to \infty$. Comparison with $s_n^{-2} \sim (\pi n/M)^8$ yields $\zeta_n \sim \pi n$. Equation (B.3) then gives the variance asymptotic in Equation (4.6).

□

D Proofs for cohort effects with balance-sheet costs

First, suppose $\kappa_r = \bar{r} = 0$ and evaluate prices at $r = 0$. Proposition 1 gives

$$\lim_{\sigma \downarrow 0} \frac{-P_c^{(1)}(0, q, \tau)}{\sigma^2 P_0(0, \tau)} = \int_0^M H_{0,0}(\tau, u) q(du).$$

The integral representation of $P^{(1)}_{\lambda,c}$ in Equation (A.4) gives

$$\lim_{\sigma \downarrow 0} \frac{-P^{(1)}_{\lambda,c}(0, q, \tau)}{P_0(0, \tau)} = \int_0^M \left[\int_0^{\min(\tau,u)} 1\, ds\right] q(du) = \int_0^M \min(\tau, u) q(du).$$

The Gaussian moment argument preceding Equation (A.5) in the proof of Lemma A.1 also justifies passage of the limit through the integrals. Combining the two contributions in the first-order log-price approximation gives

$$a\frac{P_c^{(1)}}{P_0} + \lambda\frac{P^{(1)}_{\lambda,c}}{P_0} \approx -\int_0^M \left[a\sigma^2 H_{0,0}(\tau, u) + \lambda \min(\tau, u)\right] q(du),$$

which yields Equation (4.7).

The operator J_0 defined in Online Appendix B satisfies

$$(J_0 J_0^* f)(\tau) = \int_0^M \min(\tau, u) f(u) du, \qquad \text{so} \quad \widetilde{\mathcal{G}}_{a,\lambda} = a\sigma^2 \mathcal{G}_0 + \lambda J_0 J_0^*.$$

Lemma A.2 establishes compactness and strict positivity of $\mathcal{G}_0$. The continuous symmetric

kernel of $J_0 J_0^*$ gives compactness and self-adjointness, and $\langle f, J_0 J_0^* f\rangle = \|J_0^* f\|^2 > 0$ for $f \neq 0$ because J_0^* is injective. Their nonnegative combination is therefore compact, self-adjoint, and strictly positive when the coefficients are not both zero. Moreover, the forms of the two kernels imply that its eigenfunctions are continuously differentiable and vanish at zero. Equations (3.3)–(3.5) then apply with $\mathcal{G}_{\kappa_r}$ replaced by $\widetilde{\mathcal{G}}_{a,\lambda}$, establishing Equation (4.8).

Building on these results, we extend Theorems 1 and 2 as follows.

Proposition 4 (Persistence with balance-sheet costs). *For the combined kernel in Equation (4.7), define*

$$\omega_n = \frac{a\sigma^2 \langle \widetilde{\phi}_n, \mathcal{G}_0 \widetilde{\phi}_n\rangle}{\widetilde{\eta}_n} \in [0,1].$$

Then, for every n,

$$\widetilde{\rho}_n'(0+) = -\frac{\widetilde{\phi}_n(M)^2}{2} = -\frac{1+\omega_n}{M} \in \left[-\frac{2}{M}, -\frac{1}{M}\right]. \tag{4.9}$$

The slope equals $-2/M$ when $\lambda = 0$ and $-1/M$ when $a = 0$.

Proof. Equation (3.7) gives $\widetilde{\rho}_n'(0+) = -\widetilde{\phi}_n(M)^2/2$. Since $\widetilde{\mathcal{G}}_{a,\lambda}\widetilde{\phi}_n = \widetilde{\eta}_n \widetilde{\phi}_n$ and $\widetilde{\phi}_n$ is a unit eigenfunction,

$$\widetilde{\eta}_n = a\sigma^2 \langle \widetilde{\phi}_n, \mathcal{G}_0 \widetilde{\phi}_n\rangle + \lambda \langle \widetilde{\phi}_n, J_0 J_0^* \widetilde{\phi}_n\rangle.$$

Both terms are nonnegative, so $0 \leq \omega_n \leq 1$.

By Equation (3.11), $H_{0,0}$ is homogeneous of degree three, while $\min(\tau, u)$ is homogeneous of degree one. Hence, almost everywhere,

$$(\tau \partial_\tau + u \partial_u) \widetilde{H}_{a,\lambda}(\tau, u) = 3a\sigma^2 H_{0,0}(\tau, u) + \lambda \min(\tau, u).$$

Multiply both sides of the above equation by $\widetilde{\phi}_n(\tau)\widetilde{\phi}_n(u)$ and integrate over $(0, M)^2$. The eigenfunction equation $\int_0^M \widetilde{H}_{a,\lambda}(\tau, u)\widetilde{\phi}_n(u)du = \widetilde{\eta}_n \widetilde{\phi}_n(\tau)$ then gives

$$\begin{aligned} 2\widetilde{\eta}_n \int_0^M u \widetilde{\phi}_n'(u)\widetilde{\phi}_n(u)du &= 3a\sigma^2 \langle \widetilde{\phi}_n, \mathcal{G}_0 \widetilde{\phi}_n\rangle + \lambda \langle \widetilde{\phi}_n, J_0 J_0^* \widetilde{\phi}_n\rangle \\ &= \widetilde{\eta}_n (1 + 2\omega_n). \end{aligned}$$

Integration by parts makes the left side $\widetilde{\eta}_n[M\widetilde{\phi}_n(M)^2 - 1]$. Thus $M\widetilde{\phi}_n(M)^2 = 2 + 2\omega_n$, proving Equation (4.9). The two endpoint cases follow from $\omega_n = 1$ at $\lambda = 0$ and $\omega_n = 0$ at $a = 0$. □

Proposition 5 (Variances of the principal components with balance-sheet costs). *For fixed $a\sigma^2 \geq 0$, $\lambda > 0$, and M, the combined kernel in Equation (4.7) gives*

$$\widetilde{\eta}_n \sim \frac{\lambda M^2}{[(n-\frac{1}{2})\pi]^2}, \qquad \textit{and} \quad \operatorname{Var}(\widetilde{F}_{n,t}) \sim \frac{\nu \mathbb{E}[A_i^2]\lambda^2 M^4}{[(n-\frac{1}{2})\pi]^4}, \qquad \textit{as } n \to \infty. \tag{4.10}$$

Proof. If $g = J_0 J_0^* f$, then $g'' = -f$, $g(0) = 0$, and $g'(M) = 0$. The eigenfunction equation $J_0 J_0^* f = \eta^J f$ therefore reduces to $f'' = -f/\eta^J$ with $f(0) = f'(M) = 0$. Solving this differential equation for f subject to the boundary conditions gives the decreasing eigenvalues of $J_0 J_0^*$:

$$\eta_n^J = \frac{M^2}{[(n-\frac{1}{2})\pi]^2}. \tag{D.1}$$

By the min–max principle, the eigenvalues of $\widetilde{\mathcal{G}}_{a,\lambda} = a\sigma^2 \mathcal{G}_0 + \lambda J_0 J_0^*$ satisfy the following bounds for $1 \leq k \leq n$:

$$\lambda \eta_n^J \leq \widetilde{\eta}_n \leq \lambda \eta_{n-k+1}^J + a\sigma^2 \eta_k. \tag{D.2}$$

For the lower bound, adding the nonnegative operator $a\sigma^2 \mathcal{G}_0$ cannot reduce any ordered eigenvalue. Thus, $\widetilde{\eta}_n \geq \lambda \eta_n^J$ holds.

We then apply the min–max principle to obtain the upper bound. This principle characterizes the nth eigenvalue by choosing $n-1$ directions to exclude and minimizing the largest remaining value of the quadratic form:

$$\widetilde{\eta}_n = \min_{\substack{S \subset L^2(0,M) \\ \dim S = n-1}} \max_{\substack{\|f\|=1 \\ f \perp S}} \langle f, \widetilde{\mathcal{G}}_{a,\lambda} f \rangle,$$

where S is a linear subspace. Choose an integer $1 \leq k \leq n$ and let S contain the first $n-k$ eigenfunctions of $J_0 J_0^*$ and the first $k-1$ eigenfunctions of $\mathcal{G}_0$. Their span has dimension at most $n-1$, so S can be chosen with dimension $n-1$. Every unit function $f \perp S$ then satisfies

$$\langle f, J_0 J_0^* f \rangle \leq \eta_{n-k+1}^J, \qquad \text{and} \quad \langle f, \mathcal{G}_0 f \rangle \leq \eta_k.$$

The largest remaining value of the combined quadratic form is therefore at most $\lambda \eta_{n-k+1}^J + a\sigma^2 \eta_k$. The min–max principle thus gives the stated upper bound.

Choose $k = \lfloor n^{3/4} \rfloor$. Then $k/n \to 0$ and, by Equation (D.1), $\eta_{n-k+1}^J / \eta_n^J \to 1$. Theorem 2 gives the eigenvalues $\eta_k = M^4/\zeta_k^4 = O(k^{-4})$ of $\mathcal{G}_0$, so $\eta_k = O(n^{-3}) = o(\eta_n^J)$ by Equation (D.1). Dividing Equation (D.2) by $\lambda \eta_n^J$ proves $\widetilde{\eta}_n \sim \lambda \eta_n^J = \lambda M^2/[(n-\frac{1}{2})\pi]^2$. Equation (4.8) gives the variance asymptotic. □

E Numerical solution and covariance calculation

E.1 Equations for the coefficient functions

Before deriving equilibrium bond prices, I obtain several preliminary expressions. Under Equation (5.5), an issuance (L, A) changes the bond price by the proportion

$$\Delta_\tau(A, L) = \exp[-AK_{a,\lambda}(\tau, L)] - 1.$$

The instantaneous covariance of returns and the convexity correction in their drift are

$$\begin{aligned}
\Omega(\tau, u) &= \frac{1}{dt}\operatorname{Cov}_t\left(\frac{dP_{a,\lambda}(\tau)}{P_{a,\lambda}(\tau)}, \frac{dP_{a,\lambda}(u)}{P_{a,\lambda}(u)}\right) \\
&= \sigma^2 \boldsymbol{B}_{a,\lambda}(\tau)^\top \boldsymbol{B}_{a,\lambda}(u) + \nu\mathbb{E}[\Delta_\tau \Delta_u], \\
\text{and} \quad R(\tau) &= \frac{\mathbb{E}_t[dP_{a,\lambda}(\tau)]}{P_{a,\lambda}(\tau)dt} - \frac{\mathbb{E}_t[d\log P_{a,\lambda}(\tau)]}{dt} \\
&= \frac{\sigma^2}{2}\|\boldsymbol{B}_{a,\lambda}(\tau)\|^2 + \nu\mathbb{E}[\Delta_\tau + AK_{a,\lambda}(\tau, L)].
\end{aligned} \tag{E.1}$$

Finally, set $d_0(u) = b(u, 0) - \alpha(u)A_{a,\lambda}(u)$ and $\boldsymbol{d}_a(u) = -\alpha(u)\boldsymbol{B}_{a,\lambda}(u) + \partial_\beta b(u, \beta)\boldsymbol{e}_2$, where $\boldsymbol{e}_i$ is the ith coordinate vector. Then

$$D_a(u) = d_0(u) + \boldsymbol{d}_a(u)^\top \boldsymbol{z} - \alpha(u)\langle q, K_{a,\lambda}(u, \cdot)\rangle \tag{E.2}$$

by Equation (5.5).

To derive equilibrium bond prices, I rewrite the first-order condition (2.7) in terms of returns:

$$\frac{\mathbb{E}_t[dP_{a,\lambda}(\tau)]}{P_{a,\lambda}(\tau)dt} = r + \int_0^M [a\Omega(\tau, u) + \lambda]P_{a,\lambda}(u)\Theta_a(du).$$

Substituting Equation (E.2) into Equation (5.4) and then into the above equation gives

$$\begin{aligned}
\frac{\mathbb{E}_t[dP_{a,\lambda}(\tau)]}{P_{a,\lambda}(\tau)dt} = r &+ \int_0^M [a\Omega(\tau, u) + \lambda]e^{-A_{a,\lambda}(u)}q(du) \\
&+ \int_0^M [a\Omega(\tau, u) + \lambda]\Big[d_0(u) + \boldsymbol{d}_a(u)^\top \boldsymbol{z} \\
&\qquad - \alpha(u)\langle q, K_{a,\lambda}(u, \cdot)\rangle\Big]du.
\end{aligned}$$

To apply the method of undetermined coefficients, I derive another expression for the left-hand side. Using Equation (5.5) to compute $\mathbb{E}_t[d\log P_{a,\lambda}(\tau)]/dt$ and adding the convexity correction $R(\tau)$ gives

$$\frac{\mathbb{E}_t[dP_{a,\lambda}(\tau)]}{P_{a,\lambda}(\tau)dt} = A'_{a,\lambda}(\tau) + R(\tau) + \left[\boldsymbol{B}'_{a,\lambda}(\tau) + \boldsymbol{\Lambda}\boldsymbol{B}_{a,\lambda}(\tau)\right]^\top \boldsymbol{z} + \int_0^M \left[\partial_\tau K_{a,\lambda}(\tau,u) + \partial_u K_{a,\lambda}(\tau,u)\right] q(du).$$

Here $\boldsymbol{\Lambda} = \mathrm{diag}(\kappa_r, \kappa_\beta)$. Matching constant, macro-factor, and supply terms in these two expressions gives

$$A'_{a,\lambda}(\tau) = \bar{r} - R(\tau) + \int_0^M [a\Omega(\tau,u) + \lambda] d_0(u) du, \tag{E.3}$$

$$\boldsymbol{B}'_{a,\lambda}(\tau) = \boldsymbol{e}_1 - \boldsymbol{\Lambda}\boldsymbol{B}_{a,\lambda}(\tau) + \int_0^M [a\Omega(\tau,u) + \lambda] \boldsymbol{d}_a(u) du, \tag{E.4}$$

$$\text{and} \quad (\partial_\tau + \partial_u) K_{a,\lambda}(\tau,u) = [a\Omega(\tau,u) + \lambda] e^{-A_{a,\lambda}(u)} - \int_0^M [a\Omega(\tau,v) + \lambda] \alpha(v) K_{a,\lambda}(v,u) dv. \tag{E.5}$$

E.2 Numerical solution and assessment of pricing accuracy

To obtain the coefficient functions from Equations (E.3)–(E.5), I first treat their values at Gauss–Legendre nodes as unknowns, using 32 nodes in each dimension: the maturity of the priced bond and the remaining maturity of the issuance cohorts. For any trial values at these nodes, I construct polynomial interpolants satisfying the zero-maturity boundary conditions and use them to evaluate the derivatives and integrals in the equations. This gives the equation residuals at the nodes. At each candidate parameter vector, Newton–Krylov iteration updates the unknown node values until these residuals are within the numerical tolerance, thereby determining the coefficient functions.

Specifically, a solution without stochastic issuance supplies the initial guess. Given values of $\boldsymbol{B}_{a,\lambda}$ and $K_{a,\lambda}$, Ω and R are fixed, so Equation (E.3) is linear in $A_{a,\lambda}$. I compute $A_{a,\lambda}$ by solving the corresponding linear system and then evaluate the residuals of Equations (E.4) and (E.5). To aid convergence, I transform these residuals into approximate errors in the node values of $\boldsymbol{B}_{a,\lambda}$ and $K_{a,\lambda}$ by solving a Sylvester equation based on Equations (E.4)–(E.5). Newton–Krylov iteration updates the node values of $\boldsymbol{B}_{a,\lambda}$ and $K_{a,\lambda}$ to reduce these errors. Once calibration converges, I refine to 384 nodes in each maturity dimension to compute the eleven targeted model moments more accurately, keeping the calibrated parameters fixed.

For an independent numerical check, I keep arbitrageurs' dollar holdings and their exposures to shocks fixed at the values implied by the solved prices at each state. I then solve the pricing equations as an initial-value problem in bond maturity, using 512 supply nodes. I assess numerical accuracy using the RMS differences between yields from the original and revalued prices under the stationary distribution, at annual maturities from one through

thirty years. The maximum RMS difference in yields is 0.053 bp, at the one-year maturity.

I perform two additional checks of numerical accuracy. First, I compute the RMS difference between the two sides of Equation (2.7) after dividing by the bond price. The maximum RMS error in Equation (2.7) is 0.801 bp/year. Second, I assess the stability of the variances of higher PCs under revaluation. The sum of the variances of PC3–PC30 changes by less than 0.001%, and the individual variances of PC3–PC9 change by at most 0.004%.

E.3 Covariances with and without the cohort effect

Taking $-\log P_{a,\lambda}/\tau$ in Equation (5.5) expresses each yield as a constant plus terms linear in $(r_t - \bar{r}, \beta_t)$ and q_t. These three state processes are independent, so their contributions to yield covariance add. Under the stationary distributions, Equations (2.3) and (2.4) give

$$\mathrm{Cov}(r_t, r_{t+h}) = \frac{\sigma^2}{2\kappa_r} e^{-\kappa_r h}, \qquad \text{and} \quad \mathrm{Cov}(\beta_t, \beta_{t+h}) = \frac{\sigma^2}{2\kappa_\beta} e^{-\kappa_\beta h}.$$

For the heterogeneous supply in Equation (5.3), the compensated-Poisson identity used in Equation (3.1) and the aging relation (2.2) give, for $0 \le h \le M$,

$$\mathrm{Cov}(\langle q_t, f\rangle, \langle q_{t+h}, g\rangle) = \int_h^M m_2(u) f(u) g(u-h) du.$$

Substituting these covariance formulas into the covariance of the two yields from Equation (5.5), with $f = K_{a,\lambda}(\tau_i, \cdot)$ and $g = K_{a,\lambda}(\tau_j, \cdot)$, yields

$$\begin{aligned}\Gamma_{ij}(h) = &\sum_{\ell \in \{r,\beta\}} \frac{\sigma^2}{2\kappa_\ell \tau_i \tau_j} e^{-\kappa_\ell h} B_{a,\lambda,\ell}(\tau_i) B_{a,\lambda,\ell}(\tau_j) \\ &+ \frac{1}{\tau_i \tau_j} \int_h^M m_2(u) K_{a,\lambda}(\tau_i, u) K_{a,\lambda}(\tau_j, u-h) du. \end{aligned} \tag{E.6}$$

To remove the cohort effect, I replace future supply by an independent draw from the same stationary distribution. The contribution of supply to $\boldsymbol{\Gamma}(h)$ is then zero for $h > 0$, while the covariance matrix of yields $\boldsymbol{C}$ and the contributions of the short rate and demand factor remain unchanged. I therefore omit the second term of Equation (E.6) for $h > 0$ in the case without the cohort effect. The parameters and price function are held fixed; a separate equilibrium is not estimated.